

Klasifikácia pomocou MLP

```
import torch
import torch.nn as nn
import torch.optim as optim
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import scipy.stats as stats
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from torch.utils.data import DataLoader
import torchvision
import torchvision.transforms.functional as tvf
import torch.nn.functional as tnf
from tqdm import tqdm
from argparse import Namespace
import wandb
```

Úvodné zobrazenie

```
filename = "speeddating.csv"
df = pd.read_csv(filename)
pd.set_option('display.max_columns', None)
df.head()
```

```
<ipython-input-2-7d8ac704e976>:2: DtypeWarning: Columns (3,10,11,15,16,17,18,19,39,40,41,42,43,44,51,52,53,54,55,73,74,75,76,77,78,79,80)
df = pd.read_csv(filename)
```

	has_null	wave	gender	age	age_o	d_age	d_d_age	race	race_o	samrace	importance_same_race	importance_same_
0	0	1	female	21	27	6	[4-6]	'Asian/Pacific Islander/Asian-American'	European/Caucasian-American	0		2
1	0	1	female	21	22	1	[0-1]	'Asian/Pacific Islander/Asian-American'	European/Caucasian-American	0		2
2	1	1	female	21	22	1	[0-1]	'Asian/Pacific Islander/Asian-American'	'Asian/Pacific Islander/Asian-American'	1		2
3	0	1	female	21	23	2	[2-3]	'Asian/Pacific Islander/Asian-American'	European/Caucasian-American	0		2
4	0	1	female	21	24	3	[2-3]	'Asian/Pacific Islander/Asian-American'	'Latino/Hispanic American'	0		2

Veľkosť dát (Riadky, stĺpce)

```
df.shape
```

```
(8378, 123)
```

Odstránenie prázdnych hodnôt typu "?" typom NA pre jednoduchšie spracovanie

```
df.replace(["?"], pd.NA, inplace=True)
missing_values_per_column = df.isna().sum().sum()
pd.set_option('display.max_rows', None)
print("All missing values ",missing_values_per_column)
objects = 0
for col in df.columns:
    if df[col].dtype == 'object':
        objects += 1
```

```
print("Columns of type object: ",objects)
```

```
All missing values 18372
Columns of type object: 116
```

Odstránenie stĺpcov s veľkou mierou chýbania dát

```
threshold = 0.2 # threshold for missing data
missing_percent = df.isnull().mean()
cols_to_drop = missing_percent[missing_percent > threshold].index
df = df.drop(columns=cols_to_drop)
df.shape
```

```
(8378, 122)
```

Nahradenie chýbajúcich dát

```
for column in df.columns:
    if df[column].dtype == 'object': # For categorical columns
        mode_value = df[column].mode()[0]
        df[column].fillna(mode_value, inplace=True)
    else: # For numerical columns
        median_value = df[column].median()
        df[column].fillna(median_value, inplace=True)
missing_values_per_column = df.isna().sum().sum()
print("Missing values ",missing_values_per_column)
```

<ipython-input-6-0df75afe9a00>:7: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values is a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value, inplace=True)

```
df[column].fillna(median_value, inplace=True)
<ipython-input-6-0df75afe9a00>:4: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values is a copy.
```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value, inplace=True)

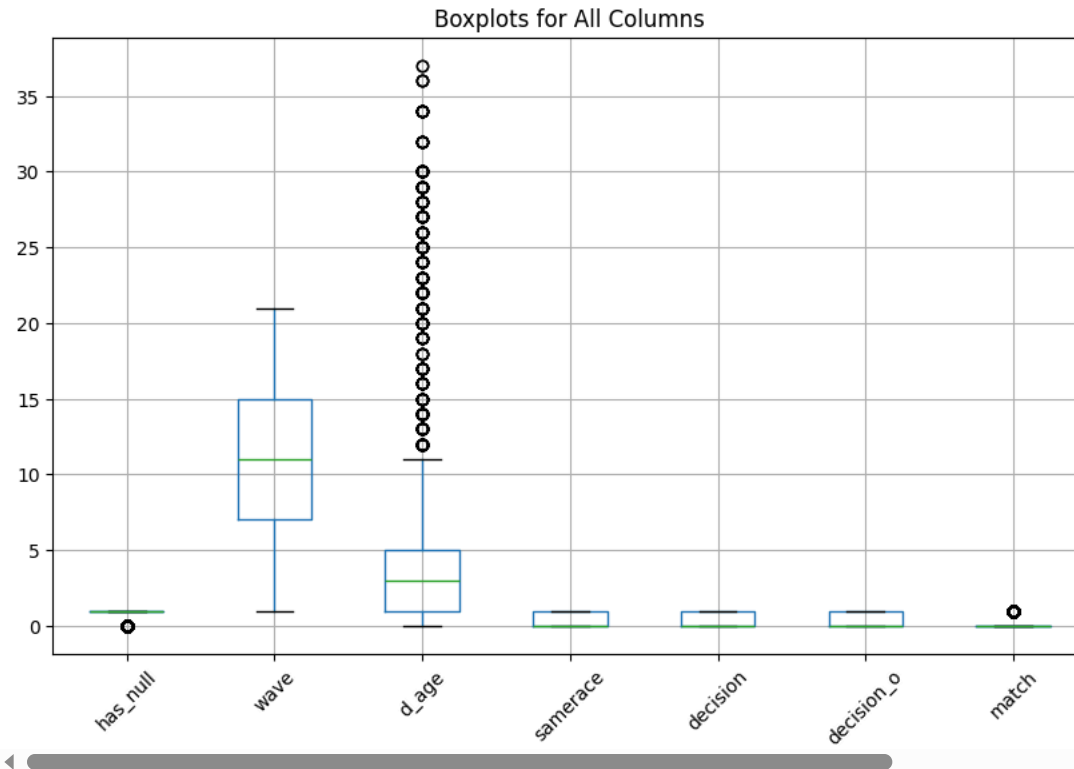
```
df[column].fillna(mode_value, inplace=True)
Missing values 0
```

```
def create_boxplots(df):
    plt.figure(figsize=(10, 6))

    # Boxplots for all numeric columns in the dataframe
    df.boxplot()

    plt.title("Boxplots for All Columns")
    plt.xticks(rotation=45)
    plt.show()

create_boxplots(df)
```



Úprava pomiešaných dátových typov na jednotný tvar

```
# Identify columns with mixed types (numbers and strings)
for column in df.columns:
    if df[column].apply(lambda x: isinstance(x, str)).sum() > 0 and df[column].apply(lambda x: isinstance(x, (int, float))).sum() > 0:
        print(f"Column '{column}' contains mixed types (both strings and numbers)")

# Convert columns with mixed types to numeric
for column in df.columns:
    if df[column].dtype == 'object': # Check if the column is object type
        def convert_to_numeric_or_keep_string(val):
            try:
                return pd.to_numeric(val) # Try to convert to numeric
            except ValueError:
                return val

        df[column] = df[column].apply(convert_to_numeric_or_keep_string) # Apply conversion function
for column in df.columns:
    df[column] = df[column].apply(lambda x: int(x) if isinstance(x, float) and x.is_integer() else x)

# Check how many missing values exist
missing_values_per_column = df.isna().sum().sum()
print("All missing values:", missing_values_per_column)
```



```
Column 'age' contains mixed types (both strings and numbers)
Column 'importance_same_race' contains mixed types (both strings and numbers)
Column 'importance_same_religion' contains mixed types (both strings and numbers)
Column 'pref_o_attractive' contains mixed types (both strings and numbers)
Column 'pref_o_sincere' contains mixed types (both strings and numbers)
Column 'pref_o_intelligence' contains mixed types (both strings and numbers)
Column 'pref_o_funny' contains mixed types (both strings and numbers)
Column 'pref_o_ambitious' contains mixed types (both strings and numbers)
Column 'attractive_important' contains mixed types (both strings and numbers)
Column 'sincere_important' contains mixed types (both strings and numbers)
Column 'intelligence_important' contains mixed types (both strings and numbers)
Column 'funny_important' contains mixed types (both strings and numbers)
Column 'ambition_important' contains mixed types (both strings and numbers)
Column 'shared_interests_important' contains mixed types (both strings and numbers)
Column 'attractive' contains mixed types (both strings and numbers)
Column 'sincere' contains mixed types (both strings and numbers)
Column 'intelligence' contains mixed types (both strings and numbers)
Column 'funny' contains mixed types (both strings and numbers)
Column 'ambition' contains mixed types (both strings and numbers)
Column 'sports' contains mixed types (both strings and numbers)
Column 'tv_sports' contains mixed types (both strings and numbers)
Column 'exercise' contains mixed types (both strings and numbers)
```

```

Column 'dining' contains mixed types (both strings and numbers)
Column 'museums' contains mixed types (both strings and numbers)
Column 'art' contains mixed types (both strings and numbers)
Column 'hiking' contains mixed types (both strings and numbers)
Column 'gaming' contains mixed types (both strings and numbers)
Column 'clubbing' contains mixed types (both strings and numbers)
Column 'reading' contains mixed types (both strings and numbers)
Column 'tv' contains mixed types (both strings and numbers)
Column 'theater' contains mixed types (both strings and numbers)
Column 'movies' contains mixed types (both strings and numbers)
Column 'concerts' contains mixed types (both strings and numbers)
Column 'music' contains mixed types (both strings and numbers)
Column 'shopping' contains mixed types (both strings and numbers)
Column 'yoga' contains mixed types (both strings and numbers)
Column 'interests_correlate' contains mixed types (both strings and numbers)
Column 'expected_happy_with_sd_people' contains mixed types (both strings and numbers)
All missing values: 0

```

Úprava kategorických dát na číselné vyjadrenie

```

df.head()
print("ok")
categorical_columns = df.select_dtypes(include=['object']).columns
for col in categorical_columns:
    le = LabelEncoder() # Encoding categorical data
    df[col] = le.fit_transform(df[col])
df.head()

```

ok

	has_null	wave	gender	age	age_o	d_age	d_d_age	race	race_o	samerace	importance_same_race	importance_same_religion	d_importan
0	0	1	0	21	27	6	2	0	3	0	2		4
1	0	1	0	21	22	1	0	0	3	0	2		4
2	1	1	0	21	22	1	0	0	0	1	2		4
3	0	1	0	21	23	2	1	0	3	0	2		4
4	0	1	0	21	24	3	1	0	2	0	2		4

Zobrazenie rozdelenia dát

```

columns = [
    'pref_o_attractive', 'pref_o_sincere', 'pref_o_intelligence', 'pref_o_funny',
    'pref_o_ambitious', 'pref_o_shared_interests', 'attractive_important',
    'sincere_important', 'intelligence_important', 'funny_important',
    'ambtition_important', 'shared_interests_important'
]

```

```

# Create a subset of the DataFrame with the specified columns
df_subset = df[columns]

```

```
sns.set(style="whitegrid")
```

```

plt.figure(figsize=(12, 8))
sns.boxplot(data=df_subset)

```

```

# Rotate the x-axis labels for better readability
plt.xticks(rotation=90)

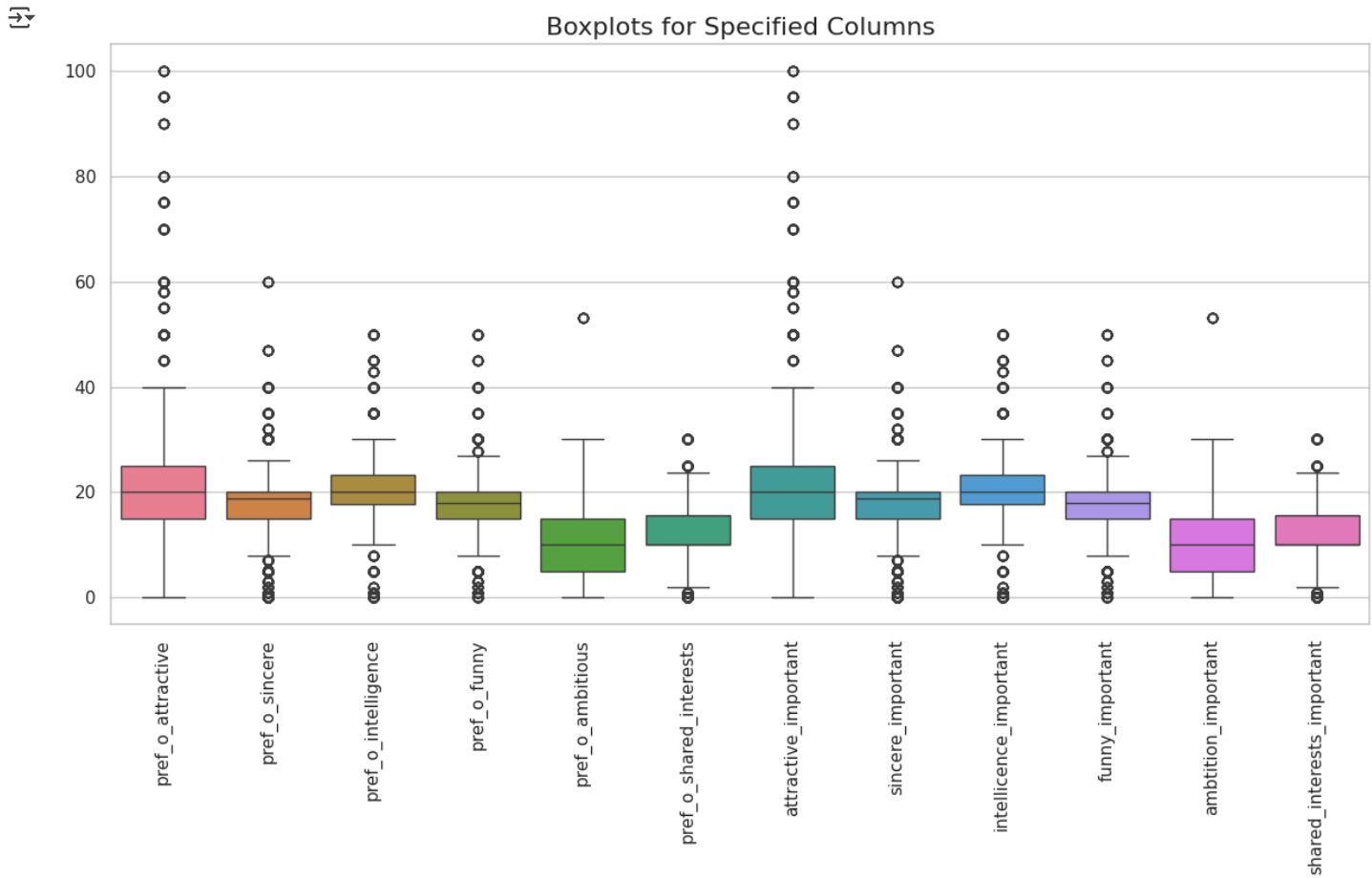
```

```
plt.title('Boxplots for Specified Columns', fontsize=16)
```

```

plt.tight_layout()
plt.show()

```



Spracovanie outlierov-odstránenie na základe kvantilov

```
columns_to_filter = [
    'pref_o_attractive', 'pref_o_sincere', 'pref_o_intelligence', 'pref_o_funny', 'pref_o_ambitious',
    'pref_o_shared_interests', 'attractive_important', 'sincere_important', 'intelligence_important',
    'funny_important', 'ambition_important', 'shared_interests_important'
]
```

```
lower_1st = df[columns_to_filter].quantile(0.01)
upper_99th = df[columns_to_filter].quantile(0.99)
```

```
outlier_mask = (df[columns_to_filter] < lower_1st) | (df[columns_to_filter] > upper_99th)
```

```
outliers_count = outlier_mask.sum()
```

```
print("Outliers removed per column:")
print(outliers_count)
```

```
df_filtered = df[~outlier_mask.any(axis=1)]
```

```
rows_removed = df.shape[0] - df_filtered.shape[0]
print(f"Total rows removed: {rows_removed}")
```

```
df_filtered = df_filtered.dropna(subset=columns_to_filter)
```

```
print(f"Final dataset size: {df_filtered.shape[0]} rows")
```

```
Outliers removed per column:
pref_o_attractive      114
pref_o_sincere         23
pref_o_intelligence    162
pref_o_funny          147
pref_o_ambitious       65
```

```
pref_o_shared_interests      0
attractive_important          77
sincere_important             23
intelligence_important        162
funny_important               147
ambtition_important           65
shared_interests_important    0
dtype: int64
Total rows removed: 738
Final dataset size: 7640 rows
```

```
columns = [
    'pref_o_attractive', 'pref_o_sincere', 'pref_o_intelligence', 'pref_o_funny',
    'pref_o_ambitious', 'pref_o_shared_interests', 'attractive_important',
    'sincere_important', 'intelligence_important', 'funny_important',
    'ambtition_important', 'shared_interests_important'
]
```

```
# Create a subset of the DataFrame with the specified columns
df_subset = df_filtered[columns]
```

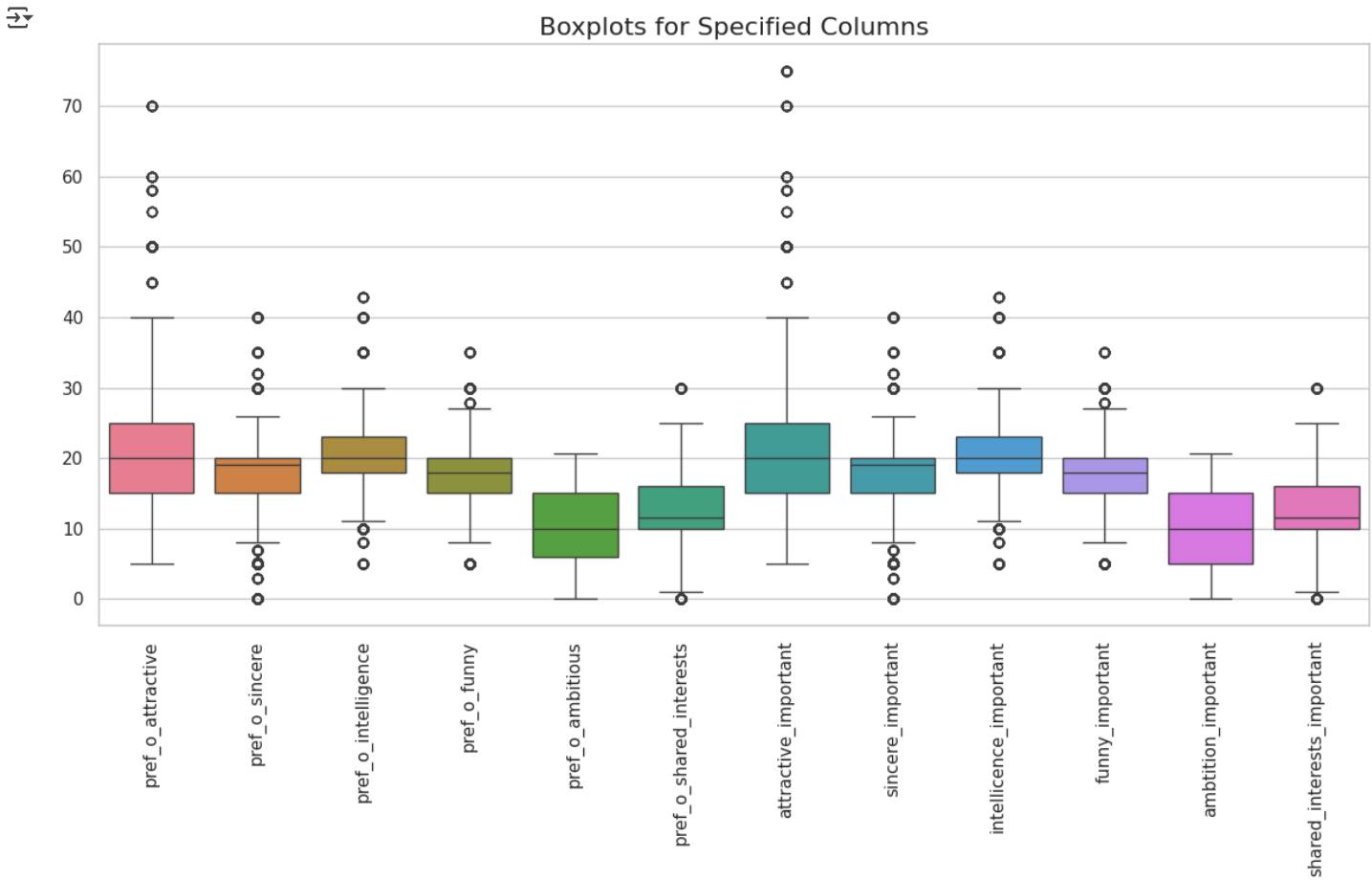
```
# Set the plot style
sns.set(style="whitegrid")
```

```
# Create the boxplots
plt.figure(figsize=(12, 8))
sns.boxplot(data=df_subset)
```

```
# Rotate the x-axis labels for better readability
plt.xticks(rotation=90)
```

```
# Add a title
plt.title('Boxplots for Specified Columns', fontsize=16)
```

```
# Display the plot
plt.tight_layout()
plt.show()
```



```
df_filtered.shape
```

```
(7640, 122)
```

Vybrali sme quantile 20 a 80 z dôvodu že upravené dáta su viac ako v polovičnom množstve oproti 25/75 z dôvodu aby sme nerobili príliš veľké zmeny v dátach, ktoré môžu byť nežiadúce

```
def clip_outliers(df_l, columns):
    # Create a copy of the DataFrame to avoid modifying the original
    df_copy = df_l.copy()

    Q1 = df_copy[columns].quantile(0.20)
    Q3 = df_copy[columns].quantile(0.8)
    IQR = Q3 - Q1

    total_clipped = 0

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    # Identify lower and upper outliers
    lower_outliers = df_copy[columns] < lower_bound
    upper_outliers = df_copy[columns] > upper_bound

    # Print outlier details for each column
    print("Clipped values per column:")
    for col in columns:
        lower_count = lower_outliers[col].sum()
        upper_count = upper_outliers[col].sum()
        total_count = lower_count + upper_count
        total_clipped += total_count # Add to the total clipped counter
        print(f"{col}: {total_count} values clipped (Lower: {lower_count}, Upper: {upper_count})")
```

```

print(f"\nTotal clipped values across all columns: {total_clipped}")

# Clip the values in the copy
df_copy[columns] = df_copy[columns].clip(lower=lower_bound, upper=upper_bound, axis=1)

return df_copy

# List of selected columns
selected_columns = [
    'pref_o_attractive', 'pref_o_sincere', 'pref_o_intelligence', 'pref_o_funny', 'pref_o_ambitious',
    'pref_o_shared_interests', 'attractive_important', 'sincere_important', 'intelligence_important',
    'funny_important', 'ambtition_important', 'shared_interests_important'
]

# Apply clipping and print results
df_clipped = clip_outliers(df_filtered, selected_columns)

🔍 Clipped values per column:
pref_o_attractive: 332 values clipped (Lower: 0, Upper: 332)
pref_o_sincere: 607 values clipped (Lower: 168, Upper: 439)
pref_o_intelligence: 75 values clipped (Lower: 0, Upper: 75)
pref_o_funny: 305 values clipped (Lower: 0, Upper: 305)
pref_o_ambitious: 0 values clipped (Lower: 0, Upper: 0)
pref_o_shared_interests: 0 values clipped (Lower: 0, Upper: 0)
attractive_important: 369 values clipped (Lower: 0, Upper: 369)
sincere_important: 605 values clipped (Lower: 167, Upper: 438)
intelligence_important: 75 values clipped (Lower: 0, Upper: 75)
funny_important: 303 values clipped (Lower: 0, Upper: 303)
ambtition_important: 0 values clipped (Lower: 0, Upper: 0)
shared_interests_important: 0 values clipped (Lower: 0, Upper: 0)

Total clipped values across all columns: 2671

```

```
df_clipped.shape
```

```
🔍 (7640, 122)
```

Zobrazenie korelácie medzi atribútami

```

column_name = "match"
correlations_sorted = df_clipped.corr()[column_name].sort_values(ascending=False)

print(correlations_sorted)

```

```
🔍
```



```

u_pret_o_attractive      -0.012717
d_attractive_important   -0.012759
pref_o_sincere           -0.015751
sincere_important        -0.017541
tv                       -0.018603
has_null                 -0.020079
movies                   -0.020646
d_pref_o_sincere         -0.024892
d_sincere_important      -0.025204
d_shared_interests_important -0.025906
d_pref_o_shared_interests -0.026551
d_tv                     -0.027901
d_movies                 -0.029119
d_importance_same_religion -0.029410
age_o                    -0.033098
age                      -0.033293
importance_same_religion -0.035357
d_age                    -0.041016
shared_interests_important -0.043364
pref_o_shared_interests -0.043512
d_importance_same_race   -0.051705
importance_same_race     -0.051908
d_d_age                  -0.061437
Name: match, dtype: float64

```

MLP Model

```

class Statistics:
    def __init__(self):
        self.values = dict()      # dvojice k -> sum, count

    def step(self, key, value):
        sum, count = 0.0, 0.0
        if key in self.values:
            sum, count = self.values[key]
        sum += value
        count += 1.0
        self.values[key] = (sum, count)

    def get(self):
        result = dict()
        for k, (sum, count) in self.values.items():
            result[k] = float(sum/count)
        return result

    @staticmethod
    def merge(s1, s2):
        result = s1.get()
        result.update(s2.get())
        return result

```

Výber atribútov

```

correlation_threshold = 0.2 # Adjust this threshold
selected_features = df_clipped.corr()["match"].abs().sort_values(ascending=False)
selected_features = selected_features[selected_features > correlation_threshold].index.tolist()
selected_features.remove("match") # Remove target column

```

```

print("Selected features:", selected_features)
print(f"Number of selected features: {len(selected_features)}")
df_clipped = df_clipped[selected_features + ["match"]] # Keep only relevant columns

```

```

➡ Selected features: ['decision', 'decision_o', 'like', 'd_like', 'funny_o', 'funny_partner', 'd_funny_o', 'd_funny_partner', 'shared_inte
Number of selected features: 18

```

```

# in this part we can ignore columns with 2high correlation
correlation_threshold_peak = 0.3
correlation_threshold = 0.25 # Adjust this threshold
selected_features = df_clipped.corr()["match"].abs().sort_values(ascending=False)
selected_features = selected_features[selected_features > correlation_threshold]
selected_features = selected_features[selected_features < correlation_threshold_peak].index.tolist()

```

```

if "match" in selected_features:
    selected_features.remove("match")

```

```
print("Selected features:", selected_features)
print(f"Number of selected features: {len(selected_features)}")
df_clipped = df_clipped[selected_features + ["match"]] # Keep only relevant columns
```

Selected features: ['like', 'd_like', 'funny_o', 'funny_partner', 'd_funny_o', 'd_funny_partner', 'shared_interests_partner', 'shared_in
Number of selected features: 13

Príprava dát na tréning a testovanie

```
df = df_clipped.copy()
# Select features and target
FEATURES = selected_features # Replace with your selected features
TARGET = "match"

X = df[FEATURES].values # Convert features to numpy
y = df[TARGET].values # Target variable

# Split data
X_train, X_val, y_train, y_val = train_test_split(X, y, test_size=0.2, random_state=42)

# Normalize (important for neural networks!)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_val = scaler.transform(X_val)

# Convert to PyTorch tensors
X_train_tensor = torch.tensor(X_train, dtype=torch.float32)
y_train_tensor = torch.tensor(y_train, dtype=torch.long) # Classification = LongTensor
X_val_tensor = torch.tensor(X_val, dtype=torch.float32)
y_val_tensor = torch.tensor(y_val, dtype=torch.long)

df.shape
```

(7640, 14)

```
from torch.utils.data import Dataset, DataLoader

class SpeedDatingDataset(Dataset):
    def __init__(self, X, y):
        self.X = X
        self.y = y

    def __len__(self):
        return len(self.X)

    def __getitem__(self, idx):
        return self.X[idx], self.y[idx]

class DataModule:
    def __init__(self, cfg, X_train, y_train, X_val, y_val):
        self.cfg = cfg

        self.dataset_train = SpeedDatingDataset(X_train, y_train)
        self.dataset_val = SpeedDatingDataset(X_val, y_val)

    def setup(self):
        self.dataloader_train = DataLoader(self.dataset_train, batch_size=self.cfg.batch_size, shuffle=True)
        self.dataloader_val = DataLoader(self.dataset_val, batch_size=self.cfg.batch_size, shuffle=False)

import torch.nn as nn

class MultilayerPerceptron(nn.Module):
    def __init__(self, drop_rate, nin, nhidden, nout):
        super().__init__()
        self.flatten = nn.Flatten()
        self.fc1 = nn.Linear(nin, nhidden)
        self.relu = nn.ReLU()
        self.dropout = nn.Dropout(drop_rate) # Dropout with 30% neurons randomly disabled
        self.fc2 = nn.Linear(nhidden, nout)

    def forward(self, x):
```

```

    x = self.flatten(x)
    x = self.fc1(x)
    x = self.relu(x)
    if self.training: # Dropout is applied only during training
        x = self.dropout(x)
    x = self.fc2(x)
    return x

def on_epoch_complete(epoch,stats):
    wandb.log({"epoch" : epoch, **stats})

import torch
import torch.optim as optim
import torch.nn as nn
from tqdm import tqdm

class Trainer:
    def __init__(self, cfg, model):
        self.device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
        self.cfg = cfg
        self.model = model.to(self.device)

        self.opt = optim.Adam(self.model.parameters(), lr=cfg.learning_rate)
        self.loss_fn = nn.CrossEntropyLoss() # Classification loss
        self.best_val_acc = 0.0 # To track best validation accuracy
        self.patience = 5 # Number of epochs with no improvement before stopping (for overfitting)
        self.no_improvement_count = 0 # Counter to track the number of epochs without improvement

    def setup(self, datamodule):
        self.datamodule = datamodule
        self.datamodule.setup()

    def compute_accuracy(self, y_hat, y):
        predictions = torch.argmax(y_hat, dim=1) # Get class with highest score
        return (predictions == y).float().mean().item()

    def fit(self):
        for epoch in range(self.cfg.max_epochs):
            self.model.train()
            stats = {"loss_train": 0.0, "acc_train": 0.0}
            num_batches = len(self.datamodule.dataloader_train) # Total training batches

            # Training loop
            with tqdm(self.datamodule.dataloader_train, desc=f"Train Epoch {epoch+1}") as progress:
                for x, y in progress:
                    x, y = x.to(self.device), y.to(self.device)

                    # Forward pass
                    y_hat = self.model(x)
                    loss = self.loss_fn(y_hat, y)

                    # Backward pass
                    self.opt.zero_grad()
                    loss.backward()
                    self.opt.step()

                    # Update statistics
                    stats["loss_train"] += loss.item() / num_batches # Average loss
                    stats["acc_train"] += self.compute_accuracy(y_hat, y) / num_batches # Average accuracy
                progress.set_postfix(stats)

            wandb.log(stats)

            # Validation Phase
            self.model.eval()
            stats_val = {"loss_val": 0.0, "acc_val": 0.0}
            num_batches = len(self.datamodule.dataloader_val) # Total validation batches

            with torch.no_grad():
                with tqdm(self.datamodule.dataloader_val, desc=f"Val Epoch {epoch+1}") as progress:
                    for x, y in progress:
                        x, y = x.to(self.device), y.to(self.device)

                        # Forward pass
                        y_hat = self.model(x)
                        loss = self.loss_fn(y_hat, y)

```

```

        # Update statistics
        stats_val["loss_val"] += loss.item() / num_batches # Average loss
        stats_val["acc_val"] += self.compute_accuracy(y_hat, y) / num_batches # Average accuracy
        progress.set_postfix(stats_val)

    # Early stopping criteria: Stop if validation accuracy reaches 100% or if overfitting is detected
    val_acc = stats_val["acc_val"]
    print(f"Epoch {epoch+1}, Validation Accuracy: {val_acc:.4f}")

    wandb.log(stats_val)

    #This part of code is temporary comented so we can train on fixed number of epochs
    """
    if val_acc == 1.0:
        print("100% Validation Accuracy reached. Stopping training.")
        break # Stop if 100% accuracy is reached

    # Track best validation accuracy and check for overfitting
    if val_acc > self.best_val_acc:
        self.best_val_acc = val_acc
        self.no_improvement_count = 0 # Reset counter if there's an improvement
    else:
        self.no_improvement_count += 1

    if self.no_improvement_count >= self.patience:
        print(f"Overfitting detected. Stopping training after {self.patience} epochs without improvement.")
        break # Stop training if validation accuracy hasn't improved for `patience` epochs
    """

```

wandb.init()

➔ wandb: Using wandb-core as the SDK backend. Please refer to <https://wandb.me/wandb-core> for more information.
 wandb: WARNING If you're specifying your api key in code, ensure this code is not shared publicly.
 wandb: WARNING Consider setting the WANDB_API_KEY environment variable, or running `wandb login` from the command line.
 wandb: No netrc file found, creating one.
 wandb: Appending key for api.wandb.ai to your netrc file: /root/.netrc
 wandb: Currently logged in as: farminmansk (farminmansk-fiit-stu) to <https://api.wandb.ai>. Use `wandb login --relogin` to force relogin
 Tracking run with wandb version 0.19.8
 Run data is saved locally in /content/wandb/run-20250326_191753-eai2y4nn
 Syncing run [misty-silence-1](#) to [Weights & Biases \(docs\)](#)
 View project at <https://wandb.ai/farminmansk-fiit-stu/uncategorized>
 View run at <https://wandb.ai/farminmansk-fiit-stu/uncategorized/runs/eai2y4nn>
 Display W&B run

from argparse import Namespace

Define hyperparameters

```

cfg = Namespace(
    batch_size=24,
    num_workers=2,
    max_epochs=10,
    learning_rate=0.005,
    num_hidden=128,
    dropping = 0.3
)

# Initialize model
model = MultilayerPerceptron(
    drop_rate = cfg.dropping,
    nin=len(FEATURES), # Number of input features
    nhidden=cfg.num_hidden,
    nout=2 # Binary classification (match or no match)
)

```

run = wandb.init(project="NS_Testing", name="Best", config=cfg)

Create DataModule

datamodule = DataModule(cfg, X_train_tensor, y_train_tensor, X_val_tensor, y_val_tensor)

Train model

```

trainer = Trainer(cfg, model)
trainer.setup(datamodule)

```

```
trainer.fit()
run = wandb.finish()
```

```

Train Epoch 1: 100%|██████████| 255/255 [00:01<00:00, 148.57it/s, loss_train=0.344, acc_train=0.849]
Val Epoch 1: 100%|██████████| 64/64 [00:00<00:00, 385.82it/s, loss_val=0.336, acc_val=0.853]
Epoch 1, Validation Accuracy: 0.8525
Train Epoch 2: 100%|██████████| 255/255 [00:01<00:00, 235.58it/s, loss_train=0.334, acc_train=0.857]
Val Epoch 2: 100%|██████████| 64/64 [00:00<00:00, 391.84it/s, loss_val=0.333, acc_val=0.857]
Epoch 2, Validation Accuracy: 0.8571
Train Epoch 3: 100%|██████████| 255/255 [00:00<00:00, 255.50it/s, loss_train=0.331, acc_train=0.856]
Val Epoch 3: 100%|██████████| 64/64 [00:00<00:00, 413.46it/s, loss_val=0.328, acc_val=0.862]
Epoch 3, Validation Accuracy: 0.8623
Train Epoch 4: 100%|██████████| 255/255 [00:01<00:00, 246.02it/s, loss_train=0.328, acc_train=0.86]
Val Epoch 4: 100%|██████████| 64/64 [00:00<00:00, 355.23it/s, loss_val=0.338, acc_val=0.846]
Epoch 4, Validation Accuracy: 0.8460
Train Epoch 5: 100%|██████████| 255/255 [00:01<00:00, 216.90it/s, loss_train=0.326, acc_train=0.858]
Val Epoch 5: 100%|██████████| 64/64 [00:00<00:00, 362.70it/s, loss_val=0.337, acc_val=0.841]
Epoch 5, Validation Accuracy: 0.8408
Train Epoch 6: 100%|██████████| 255/255 [00:01<00:00, 212.04it/s, loss_train=0.327, acc_train=0.856]
Val Epoch 6: 100%|██████████| 64/64 [00:00<00:00, 419.46it/s, loss_val=0.331, acc_val=0.845]
Epoch 6, Validation Accuracy: 0.8451
Train Epoch 7: 100%|██████████| 255/255 [00:01<00:00, 226.44it/s, loss_train=0.324, acc_train=0.857]
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 359.25it/s, loss_val=0.33, acc_val=0.851]
Epoch 7, Validation Accuracy: 0.8512
Train Epoch 8: 100%|██████████| 255/255 [00:01<00:00, 206.12it/s, loss_train=0.322, acc_train=0.862]
Val Epoch 8: 100%|██████████| 64/64 [00:00<00:00, 351.37it/s, loss_val=0.327, acc_val=0.864]
Epoch 8, Validation Accuracy: 0.8636
Train Epoch 9: 100%|██████████| 255/255 [00:01<00:00, 231.78it/s, loss_train=0.322, acc_train=0.861]
Val Epoch 9: 100%|██████████| 64/64 [00:00<00:00, 389.91it/s, loss_val=0.33, acc_val=0.853]
Epoch 9, Validation Accuracy: 0.8529
Train Epoch 10: 100%|██████████| 255/255 [00:00<00:00, 270.43it/s, loss_train=0.323, acc_train=0.861]
Val Epoch 10: 100%|██████████| 64/64 [00:00<00:00, 413.06it/s, loss_val=0.33, acc_val=0.85]
Epoch 10, Validation Accuracy: 0.8503

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.86119
acc_val   0.85026
loss_train 0.32345
loss_val   0.32971

```

View run **misty-silence-1** at: <https://wandb.ai/farminmansk-fiit-stu/uncategorized/runs/eai2y4nn>

View project at: <https://wandb.ai/farminmansk-fiit-stu/uncategorized>

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_191753-eai2y4nn/logs

```
def train_model(batch_size=16, num_workers=2, max_epochs=10, learning_rate=0.1, num_hidden=128, drop_rate=0.2, project_name="NS_Testing", ru
```

```

    cfg = Namespace(
        batch_size=batch_size,
        num_workers=num_workers,
        max_epochs=max_epochs,
        learning_rate=learning_rate,
        num_hidden=num_hidden,
        drop_rate=drop_rate
    )

```

```

    model = MultilayerPerceptron(
        drop_rate=cfg.drop_rate,
        nin=len(FEATURES),
        nhidden=cfg.num_hidden,
        nout=2
    )

```

```
run = wandb.init(project=project_name, name=run_name, config=vars(cfg))
```

```
datamodule = DataModule(cfg, X_train_tensor, y_train_tensor, X_val_tensor, y_val_tensor)

trainer = Trainer(cfg, model)
trainer.setup(datamodule)
trainer.fit()

wandb.finish()
```

Testovanie s rôzne nastavenými parametrami

```
# First 12
train_model(batch_size=16, learning_rate=0.1, num_hidden=128, drop_rate=0.3, run_name="B1N1L1D1")
train_model(batch_size=16, learning_rate=0.1, num_hidden=128, drop_rate=0.2, run_name="B1N1L1D2")
train_model(batch_size=16, learning_rate=0.01, num_hidden=128, drop_rate=0.3, run_name="B1N1L2D1")
train_model(batch_size=16, learning_rate=0.01, num_hidden=128, drop_rate=0.2, run_name="B1N1L2D2")
train_model(batch_size=16, learning_rate=0.005, num_hidden=128, drop_rate=0.3, run_name="B1N1L3D1")
train_model(batch_size=16, learning_rate=0.005, num_hidden=128, drop_rate=0.2, run_name="B1N1L3D2")
train_model(batch_size=16, learning_rate=0.1, num_hidden=256, drop_rate=0.3, run_name="B1N2L1D1")
train_model(batch_size=16, learning_rate=0.1, num_hidden=256, drop_rate=0.2, run_name="B1N2L1D2")
train_model(batch_size=16, learning_rate=0.01, num_hidden=256, drop_rate=0.3, run_name="B1N2L2D1")
train_model(batch_size=16, learning_rate=0.01, num_hidden=256, drop_rate=0.2, run_name="B1N2L2D2")
train_model(batch_size=16, learning_rate=0.005, num_hidden=256, drop_rate=0.3, run_name="B1N2L3D1")
train_model(batch_size=16, learning_rate=0.005, num_hidden=256, drop_rate=0.2, run_name="B1N2L3D2")
```



Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_191835-y10av1js

Syncing run **B1N1L1D1** to [Weights & Biases \(docs\)](#)View project at https://wandb.ai/farminmansk-fiit-stu/NS_TestingView run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/y10av1js

```

Train Epoch 1: 100%|██████████| 382/382 [00:01<00:00, 267.72it/s, loss_train=1.1, acc_train=0.807]
Val Epoch 1: 100%|██████████| 96/96 [00:00<00:00, 433.24it/s, loss_val=0.565, acc_val=0.824]
Epoch 1, Validation Accuracy: 0.8242
Train Epoch 2: 100%|██████████| 382/382 [00:02<00:00, 157.15it/s, loss_train=0.533, acc_train=0.827]
Val Epoch 2: 100%|██████████| 96/96 [00:00<00:00, 194.23it/s, loss_val=1.05, acc_val=0.712]
Epoch 2, Validation Accuracy: 0.7116
Train Epoch 3: 100%|██████████| 382/382 [00:04<00:00, 76.50it/s, loss_train=0.554, acc_train=0.826]
Val Epoch 3: 100%|██████████| 96/96 [00:00<00:00, 184.70it/s, loss_val=0.532, acc_val=0.831]
Epoch 3, Validation Accuracy: 0.8314
Train Epoch 4: 100%|██████████| 382/382 [00:02<00:00, 184.20it/s, loss_train=0.524, acc_train=0.833]
Val Epoch 4: 100%|██████████| 96/96 [00:00<00:00, 430.47it/s, loss_val=0.474, acc_val=0.818]
Epoch 4, Validation Accuracy: 0.8184
Train Epoch 5: 100%|██████████| 382/382 [00:01<00:00, 243.84it/s, loss_train=0.465, acc_train=0.836]
Val Epoch 5: 100%|██████████| 96/96 [00:00<00:00, 406.43it/s, loss_val=0.437, acc_val=0.827]
Epoch 5, Validation Accuracy: 0.8268
Train Epoch 6: 100%|██████████| 382/382 [00:01<00:00, 253.62it/s, loss_train=0.51, acc_train=0.833]
Val Epoch 6: 100%|██████████| 96/96 [00:00<00:00, 447.03it/s, loss_val=0.512, acc_val=0.833]
Epoch 6, Validation Accuracy: 0.8333
Train Epoch 7: 100%|██████████| 382/382 [00:01<00:00, 251.53it/s, loss_train=0.538, acc_train=0.835]
Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 464.71it/s, loss_val=0.475, acc_val=0.834]
Epoch 7, Validation Accuracy: 0.8340
Train Epoch 8: 100%|██████████| 382/382 [00:01<00:00, 229.66it/s, loss_train=0.525, acc_train=0.831]
Val Epoch 8: 100%|██████████| 96/96 [00:00<00:00, 415.81it/s, loss_val=0.69, acc_val=0.836]
Epoch 8, Validation Accuracy: 0.8359
Train Epoch 9: 100%|██████████| 382/382 [00:01<00:00, 206.40it/s, loss_train=0.51, acc_train=0.837]
Val Epoch 9: 100%|██████████| 96/96 [00:00<00:00, 486.93it/s, loss_val=0.716, acc_val=0.833]
Epoch 9, Validation Accuracy: 0.8327
Train Epoch 10: 100%|██████████| 382/382 [00:01<00:00, 237.52it/s, loss_train=0.474, acc_train=0.84]
Val Epoch 10: 100%|██████████| 96/96 [00:00<00:00, 355.91it/s, loss_val=0.435, acc_val=0.834]
Epoch 10, Validation Accuracy: 0.8340

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.83966
acc_val   0.83398
loss_train 0.47447
loss_val   0.43513

```

View run **B1N1L1D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/y10av1jsView project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_191835-y10av1js/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_191901-htjc9oap

Syncing run **B1N1L1D2** to [Weights & Biases \(docs\)](#)View project at https://wandb.ai/farminmansk-fiit-stu/NS_TestingView run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/htjc9oap

```

Train Epoch 1: 100%|██████████| 382/382 [00:01<00:00, 248.65it/s, loss_train=1.05, acc_train=0.803]
Val Epoch 1: 100%|██████████| 96/96 [00:00<00:00, 405.49it/s, loss_val=0.815, acc_val=0.829]
Epoch 1, Validation Accuracy: 0.8294
Train Epoch 2: 100%|██████████| 382/382 [00:01<00:00, 242.25it/s, loss_train=0.545, acc_train=0.825]
Val Epoch 2: 100%|██████████| 96/96 [00:00<00:00, 421.83it/s, loss_val=0.492, acc_val=0.824]
Epoch 2, Validation Accuracy: 0.8236
Train Epoch 3: 100%|██████████| 382/382 [00:01<00:00, 251.57it/s, loss_train=0.492, acc_train=0.831]
Val Epoch 3: 100%|██████████| 96/96 [00:00<00:00, 442.17it/s, loss_val=0.528, acc_val=0.835]
Epoch 3, Validation Accuracy: 0.8346
Train Epoch 4: 100%|██████████| 382/382 [00:01<00:00, 237.64it/s, loss_train=0.454, acc_train=0.834]
Val Epoch 4: 100%|██████████| 96/96 [00:00<00:00, 435.67it/s, loss_val=0.447, acc_val=0.835]
Epoch 4, Validation Accuracy: 0.8346
Train Epoch 5: 100%|██████████| 382/382 [00:01<00:00, 234.45it/s, loss_train=0.442, acc_train=0.835]
Val Epoch 5: 100%|██████████| 96/96 [00:00<00:00, 319.00it/s, loss_val=0.597, acc_val=0.833]
Epoch 5, Validation Accuracy: 0.8327
Train Epoch 6: 100%|██████████| 382/382 [00:01<00:00, 204.96it/s, loss_train=0.498, acc_train=0.832]
Val Epoch 6: 100%|██████████| 96/96 [00:00<00:00, 415.68it/s, loss_val=0.64, acc_val=0.831]
Epoch 6, Validation Accuracy: 0.8314
Train Epoch 7: 100%|██████████| 382/382 [00:01<00:00, 208.36it/s, loss_train=0.62, acc_train=0.833]
Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 371.76it/s, loss_val=0.514, acc_val=0.831]

```

```
Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 317.20it/s, loss_val=0.624, acc_val=0.834]
Epoch 7, Validation Accuracy: 0.8307
Train Epoch 8: 100%|██████████| 382/382 [00:01<00:00, 218.75it/s, loss_train=0.521, acc_train=0.833]
Val Epoch 8: 100%|██████████| 96/96 [00:00<00:00, 444.27it/s, loss_val=0.686, acc_val=0.835]
Epoch 8, Validation Accuracy: 0.8346
Train Epoch 9: 100%|██████████| 382/382 [00:01<00:00, 247.97it/s, loss_train=0.49, acc_train=0.836]
Val Epoch 9: 100%|██████████| 96/96 [00:00<00:00, 432.10it/s, loss_val=0.567, acc_val=0.835]
Epoch 9, Validation Accuracy: 0.8346
Train Epoch 10: 100%|██████████| 382/382 [00:01<00:00, 237.03it/s, loss_train=0.482, acc_train=0.837]
Val Epoch 10: 100%|██████████| 96/96 [00:00<00:00, 435.78it/s, loss_val=0.471, acc_val=0.833]
Epoch 10, Validation Accuracy: 0.8333
```

Run history:



Run summary:

acc_train 0.83737
acc_val 0.83333
loss_train 0.48236
loss_val 0.47089

View run **B1N1L1D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/htjc9oap

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_191901-htjc9oap/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_191923-7bqrasd8

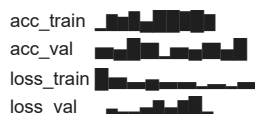
Syncing run **B1N1L2D1** to [Weights & Biases](https://wandb.ai/farminmansk-fiit-stu/NS_Testing) ([docs](#))

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/7bqrasd8

```
Train Epoch 1: 100%|██████████| 382/382 [00:01<00:00, 248.26it/s, loss_train=0.354, acc_train=0.845]
Val Epoch 1: 100%|██████████| 96/96 [00:00<00:00, 366.44it/s, loss_val=0.335, acc_val=0.847]
Epoch 1, Validation Accuracy: 0.8470
Train Epoch 2: 100%|██████████| 382/382 [00:01<00:00, 234.68it/s, loss_train=0.344, acc_train=0.851]
Val Epoch 2: 100%|██████████| 96/96 [00:00<00:00, 454.77it/s, loss_val=0.341, acc_val=0.845]
Epoch 2, Validation Accuracy: 0.8451
Train Epoch 3: 100%|██████████| 382/382 [00:01<00:00, 205.34it/s, loss_train=0.341, acc_train=0.85]
Val Epoch 3: 100%|██████████| 96/96 [00:00<00:00, 294.35it/s, loss_val=0.34, acc_val=0.853]
Epoch 3, Validation Accuracy: 0.8529
Train Epoch 4: 100%|██████████| 382/382 [00:01<00:00, 219.48it/s, loss_train=0.341, acc_train=0.853]
Val Epoch 4: 100%|██████████| 96/96 [00:00<00:00, 431.68it/s, loss_val=0.337, acc_val=0.852]
Epoch 4, Validation Accuracy: 0.8516
Train Epoch 5: 100%|██████████| 382/382 [00:01<00:00, 259.25it/s, loss_train=0.34, acc_train=0.848]
Val Epoch 5: 100%|██████████| 96/96 [00:00<00:00, 406.72it/s, loss_val=0.345, acc_val=0.839]
Epoch 5, Validation Accuracy: 0.8392
Train Epoch 6: 100%|██████████| 382/382 [00:02<00:00, 190.03it/s, loss_train=0.34, acc_train=0.854]
Val Epoch 6: 100%|██████████| 96/96 [00:00<00:00, 395.58it/s, loss_val=0.352, acc_val=0.846]
Epoch 6, Validation Accuracy: 0.8464
Train Epoch 7: 100%|██████████| 382/382 [00:01<00:00, 230.00it/s, loss_train=0.334, acc_train=0.854]
Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 407.23it/s, loss_val=0.344, acc_val=0.845]
Epoch 7, Validation Accuracy: 0.8451
Train Epoch 8: 100%|██████████| 382/382 [00:01<00:00, 251.63it/s, loss_train=0.337, acc_train=0.852]
Val Epoch 8: 100%|██████████| 96/96 [00:00<00:00, 416.58it/s, loss_val=0.351, acc_val=0.852]
Epoch 8, Validation Accuracy: 0.8516
Train Epoch 9: 100%|██████████| 382/382 [00:01<00:00, 234.34it/s, loss_train=0.333, acc_train=0.854]
Val Epoch 9: 100%|██████████| 96/96 [00:00<00:00, 363.31it/s, loss_val=0.358, acc_val=0.846]
Epoch 9, Validation Accuracy: 0.8457
Train Epoch 10: 100%|██████████| 382/382 [00:01<00:00, 239.37it/s, loss_train=0.339, acc_train=0.851]
Val Epoch 10: 100%|██████████| 96/96 [00:00<00:00, 424.63it/s, loss_val=0.338, acc_val=0.852]
Epoch 10, Validation Accuracy: 0.8522
```

Run history:



Run summary:


```
acc_train 0.85079
acc_val 0.85221
loss_train 0.33892
loss_val 0.33784
```

View run **B1N1L2D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/7bqrasd8

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_191923-7bqrasd8/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_191945-ikp50yaj

Syncing run **B1N1L2D2** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/ikp50yaj

```
Train Epoch 1: 100%|██████████| 382/382 [00:01<00:00, 244.72it/s, loss_train=0.353, acc_train=0.851]
Val Epoch 1: 100%|██████████| 96/96 [00:00<00:00, 383.03it/s, loss_val=0.333, acc_val=0.844]
Epoch 1, Validation Accuracy: 0.8444
Train Epoch 2: 100%|██████████| 382/382 [00:01<00:00, 221.19it/s, loss_train=0.34, acc_train=0.852]
Val Epoch 2: 100%|██████████| 96/96 [00:00<00:00, 317.27it/s, loss_val=0.341, acc_val=0.853]
Epoch 2, Validation Accuracy: 0.8529
Train Epoch 3: 100%|██████████| 382/382 [00:01<00:00, 200.55it/s, loss_train=0.333, acc_train=0.857]
Val Epoch 3: 100%|██████████| 96/96 [00:00<00:00, 318.64it/s, loss_val=0.333, acc_val=0.859]
Epoch 3, Validation Accuracy: 0.8594
Train Epoch 4: 100%|██████████| 382/382 [00:01<00:00, 219.42it/s, loss_train=0.338, acc_train=0.853]
Val Epoch 4: 100%|██████████| 96/96 [00:00<00:00, 417.10it/s, loss_val=0.335, acc_val=0.855]
Epoch 4, Validation Accuracy: 0.8548
Train Epoch 5: 100%|██████████| 382/382 [00:01<00:00, 226.17it/s, loss_train=0.334, acc_train=0.852]
Val Epoch 5: 100%|██████████| 96/96 [00:00<00:00, 357.21it/s, loss_val=0.339, acc_val=0.854]
Epoch 5, Validation Accuracy: 0.8542
Train Epoch 6: 100%|██████████| 382/382 [00:01<00:00, 247.04it/s, loss_train=0.33, acc_train=0.853]
Val Epoch 6: 100%|██████████| 96/96 [00:00<00:00, 405.98it/s, loss_val=0.341, acc_val=0.851]
Epoch 6, Validation Accuracy: 0.8509
Train Epoch 7: 100%|██████████| 382/382 [00:01<00:00, 220.92it/s, loss_train=0.336, acc_train=0.854]
Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 377.96it/s, loss_val=0.337, acc_val=0.855]
Epoch 7, Validation Accuracy: 0.8555
Train Epoch 8: 100%|██████████| 382/382 [00:01<00:00, 237.52it/s, loss_train=0.333, acc_train=0.854]
Val Epoch 8: 100%|██████████| 96/96 [00:00<00:00, 481.82it/s, loss_val=0.335, acc_val=0.859]
Epoch 8, Validation Accuracy: 0.8587
Train Epoch 9: 100%|██████████| 382/382 [00:01<00:00, 230.53it/s, loss_train=0.332, acc_train=0.854]
Val Epoch 9: 100%|██████████| 96/96 [00:00<00:00, 293.16it/s, loss_val=0.335, acc_val=0.856]
Epoch 9, Validation Accuracy: 0.8561
Train Epoch 10: 100%|██████████| 382/382 [00:01<00:00, 219.73it/s, loss_train=0.331, acc_train=0.857]
Val Epoch 10: 100%|██████████| 96/96 [00:00<00:00, 266.01it/s, loss_val=0.339, acc_val=0.848]
Epoch 10, Validation Accuracy: 0.8483
```

Run history:

```
acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████
```

Run summary:

```
acc_train 0.857
acc_val 0.84831
loss_train 0.33109
loss_val 0.33895
```

View run **B1N1L2D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/ikp50yaj

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_191945-ikp50yaj/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192007-9m5od2j5

Syncing run **B1N1L3D1** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/9m5od2j5

```
Train Epoch 1: 100%|██████████| 382/382 [00:01<00:00, 252.92it/s, loss_train=0.345, acc_train=0.85]
Val Epoch 1: 100%|██████████| 96/96 [00:00<00:00, 414.92it/s, loss_val=0.337, acc_val=0.85]
Epoch 1, Validation Accuracy: 0.8503
Train Epoch 2: 100%|██████████| 382/382 [00:01<00:00, 231.53it/s, loss_train=0.33, acc_train=0.854]
Val Epoch 2: 100%|██████████| 96/96 [00:00<00:00, 397.12it/s, loss_val=0.334, acc_val=0.85]
Epoch 2, Validation Accuracy: 0.8503
Train Epoch 3: 100%|██████████| 382/382 [00:01<00:00, 231.49it/s, loss_train=0.334, acc_train=0.857]
Val Epoch 3: 100%|██████████| 96/96 [00:00<00:00, 390.32it/s, loss_val=0.326, acc_val=0.853]
Epoch 3, Validation Accuracy: 0.8529
Train Epoch 4: 100%|██████████| 382/382 [00:01<00:00, 224.58it/s, loss_train=0.332, acc_train=0.855]
```

```

Val Epoch 4: 100%|██████████| 96/96 [00:00<00:00, 390.35it/s, loss_val=0.337, acc_val=0.863]
Epoch 4, Validation Accuracy: 0.8633
Train Epoch 5: 100%|██████████| 382/382 [00:01<00:00, 238.67it/s, loss_train=0.327, acc_train=0.857]
Val Epoch 5: 100%|██████████| 96/96 [00:00<00:00, 393.70it/s, loss_val=0.336, acc_val=0.854]
Epoch 5, Validation Accuracy: 0.8535
Train Epoch 6: 100%|██████████| 382/382 [00:01<00:00, 246.42it/s, loss_train=0.332, acc_train=0.856]
Val Epoch 6: 100%|██████████| 96/96 [00:00<00:00, 449.50it/s, loss_val=0.324, acc_val=0.852]
Epoch 6, Validation Accuracy: 0.8522
Train Epoch 7: 100%|██████████| 382/382 [00:01<00:00, 263.74it/s, loss_train=0.327, acc_train=0.857]
Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 426.76it/s, loss_val=0.332, acc_val=0.86]
Epoch 7, Validation Accuracy: 0.8600
Train Epoch 8: 100%|██████████| 382/382 [00:01<00:00, 220.31it/s, loss_train=0.325, acc_train=0.856]
Val Epoch 8: 100%|██████████| 96/96 [00:00<00:00, 284.67it/s, loss_val=0.332, acc_val=0.852]
Epoch 8, Validation Accuracy: 0.8516
Train Epoch 9: 100%|██████████| 382/382 [00:01<00:00, 200.78it/s, loss_train=0.322, acc_train=0.858]
Val Epoch 9: 100%|██████████| 96/96 [00:00<00:00, 280.29it/s, loss_val=0.33, acc_val=0.857]
Epoch 9, Validation Accuracy: 0.8568
Train Epoch 10: 100%|██████████| 382/382 [00:01<00:00, 220.74it/s, loss_train=0.324, acc_train=0.857]
Val Epoch 10: 100%|██████████| 96/96 [00:00<00:00, 333.49it/s, loss_val=0.326, acc_val=0.859]
Epoch 10, Validation Accuracy: 0.8594

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.85749
acc_val   0.85938
loss_train 0.32447
loss_val  0.32575

```

View run **B1N1L3D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/9m5od2j5

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192007-9m5od2j5/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192029-hiw0054p

Syncing run **B1N1L3D2** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

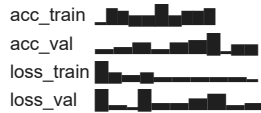
View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/hiw0054p

```

Train Epoch 1: 100%|██████████| 382/382 [00:01<00:00, 252.40it/s, loss_train=0.342, acc_train=0.853]
Val Epoch 1: 100%|██████████| 96/96 [00:00<00:00, 427.74it/s, loss_val=0.346, acc_val=0.849]
Epoch 1, Validation Accuracy: 0.8490
Train Epoch 2: 100%|██████████| 382/382 [00:01<00:00, 216.44it/s, loss_train=0.331, acc_train=0.859]
Val Epoch 2: 100%|██████████| 96/96 [00:00<00:00, 338.27it/s, loss_val=0.331, acc_val=0.85]
Epoch 2, Validation Accuracy: 0.8503
Train Epoch 3: 100%|██████████| 382/382 [00:01<00:00, 236.76it/s, loss_train=0.328, acc_train=0.858]
Val Epoch 3: 100%|██████████| 96/96 [00:00<00:00, 433.09it/s, loss_val=0.327, acc_val=0.854]
Epoch 3, Validation Accuracy: 0.8542
Train Epoch 4: 100%|██████████| 382/382 [00:01<00:00, 205.95it/s, loss_train=0.33, acc_train=0.857]
Val Epoch 4: 100%|██████████| 96/96 [00:00<00:00, 372.79it/s, loss_val=0.347, acc_val=0.85]
Epoch 4, Validation Accuracy: 0.8496
Train Epoch 5: 100%|██████████| 382/382 [00:01<00:00, 231.61it/s, loss_train=0.325, acc_train=0.856]
Val Epoch 5: 100%|██████████| 96/96 [00:00<00:00, 433.91it/s, loss_val=0.333, acc_val=0.854]
Epoch 5, Validation Accuracy: 0.8535
Train Epoch 6: 100%|██████████| 382/382 [00:02<00:00, 186.59it/s, loss_train=0.323, acc_train=0.862]
Val Epoch 6: 100%|██████████| 96/96 [00:00<00:00, 334.69it/s, loss_val=0.334, acc_val=0.855]
Epoch 6, Validation Accuracy: 0.8555
Train Epoch 7: 100%|██████████| 382/382 [00:01<00:00, 216.31it/s, loss_train=0.323, acc_train=0.857]
Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 284.95it/s, loss_val=0.338, acc_val=0.859]
Epoch 7, Validation Accuracy: 0.8587
Train Epoch 8: 100%|██████████| 382/382 [00:01<00:00, 223.47it/s, loss_train=0.325, acc_train=0.858]
Val Epoch 8: 100%|██████████| 96/96 [00:00<00:00, 360.01it/s, loss_val=0.343, acc_val=0.848]
Epoch 8, Validation Accuracy: 0.8477
Train Epoch 9: 100%|██████████| 382/382 [00:01<00:00, 219.19it/s, loss_train=0.324, acc_train=0.857]
Val Epoch 9: 100%|██████████| 96/96 [00:00<00:00, 266.96it/s, loss_val=0.329, acc_val=0.852]
Epoch 9, Validation Accuracy: 0.8522
Train Epoch 10: 100%|██████████| 382/382 [00:01<00:00, 210.30it/s, loss_train=0.321, acc_train=0.859]
Val Epoch 10: 100%|██████████| 96/96 [00:00<00:00, 441.97it/s, loss_val=0.334, acc_val=0.853]
Epoch 10, Validation Accuracy: 0.8529

```

Run history:

**Run summary:**

acc_train 0.85864
 acc_val 0.85286
 loss_train 0.32131
 loss_val 0.33405

View run **B1N1L3D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/hiw0054p

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192029-hiw0054p/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192052-v6ohwhvv

Syncing run **B1N2L1D1** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/v6ohwhvv

Train Epoch 1: 100%|██████████| 382/382 [00:01<00:00, 237.03it/s, loss_train=1.97, acc_train=0.795]
 Val Epoch 1: 100%|██████████| 96/96 [00:00<00:00, 385.78it/s, loss_val=2.25, acc_val=0.846]
 Epoch 1, Validation Accuracy: 0.8464
 Train Epoch 2: 100%|██████████| 382/382 [00:01<00:00, 197.71it/s, loss_train=1.03, acc_train=0.816]
 Val Epoch 2: 100%|██████████| 96/96 [00:00<00:00, 366.33it/s, loss_val=0.874, acc_val=0.83]
 Epoch 2, Validation Accuracy: 0.8301
 Train Epoch 3: 100%|██████████| 382/382 [00:01<00:00, 208.82it/s, loss_train=0.667, acc_train=0.823]
 Val Epoch 3: 100%|██████████| 96/96 [00:00<00:00, 284.31it/s, loss_val=0.465, acc_val=0.836]
 Epoch 3, Validation Accuracy: 0.8359
 Train Epoch 4: 100%|██████████| 382/382 [00:01<00:00, 191.50it/s, loss_train=0.586, acc_train=0.83]
 Val Epoch 4: 100%|██████████| 96/96 [00:00<00:00, 398.67it/s, loss_val=0.705, acc_val=0.829]
 Epoch 4, Validation Accuracy: 0.8288
 Train Epoch 5: 100%|██████████| 382/382 [00:01<00:00, 206.97it/s, loss_train=0.802, acc_train=0.825]
 Val Epoch 5: 100%|██████████| 96/96 [00:00<00:00, 313.71it/s, loss_val=2.83, acc_val=0.841]
 Epoch 5, Validation Accuracy: 0.8411
 Train Epoch 6: 100%|██████████| 382/382 [00:01<00:00, 217.32it/s, loss_train=0.825, acc_train=0.829]
 Val Epoch 6: 100%|██████████| 96/96 [00:00<00:00, 391.07it/s, loss_val=0.531, acc_val=0.834]
 Epoch 6, Validation Accuracy: 0.8340
 Train Epoch 7: 100%|██████████| 382/382 [00:01<00:00, 215.05it/s, loss_train=0.536, acc_train=0.833]
 Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 294.61it/s, loss_val=0.85, acc_val=0.833]
 Epoch 7, Validation Accuracy: 0.8327
 Train Epoch 8: 100%|██████████| 382/382 [00:01<00:00, 231.68it/s, loss_train=0.514, acc_train=0.835]
 Val Epoch 8: 100%|██████████| 96/96 [00:00<00:00, 493.66it/s, loss_val=0.592, acc_val=0.835]
 Epoch 8, Validation Accuracy: 0.8346
 Train Epoch 9: 100%|██████████| 382/382 [00:01<00:00, 223.80it/s, loss_train=0.615, acc_train=0.834]
 Val Epoch 9: 100%|██████████| 96/96 [00:00<00:00, 434.39it/s, loss_val=0.928, acc_val=0.83]
 Epoch 9, Validation Accuracy: 0.8301
 Train Epoch 10: 100%|██████████| 382/382 [00:01<00:00, 216.49it/s, loss_train=0.585, acc_train=0.831]
 Val Epoch 10: 100%|██████████| 96/96 [00:00<00:00, 403.46it/s, loss_val=0.696, acc_val=0.831]
 Epoch 10, Validation Accuracy: 0.8314

Run history:**Run summary:**

acc_train 0.83066
 acc_val 0.83138
 loss_train 0.58456
 loss_val 0.6961

View run **B1N2L1D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/v6ohwhvv

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192052-v6ohwhvv/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192115-e7ykh021

Syncing run **B1N2L1D2** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/e7ykh021

```

Train Epoch 1: 100%|██████████| 382/382 [00:01<00:00, 246.38it/s, loss_train=1.28, acc_train=0.807]
Val Epoch 1: 100%|██████████| 96/96 [00:00<00:00, 358.79it/s, loss_val=1.06, acc_val=0.781]
Epoch 1, Validation Accuracy: 0.7812
Train Epoch 2: 100%|██████████| 382/382 [00:01<00:00, 237.80it/s, loss_train=0.87, acc_train=0.819]
Val Epoch 2: 100%|██████████| 96/96 [00:00<00:00, 466.72it/s, loss_val=0.806, acc_val=0.77]
Epoch 2, Validation Accuracy: 0.7702
Train Epoch 3: 100%|██████████| 382/382 [00:01<00:00, 241.55it/s, loss_train=0.944, acc_train=0.821]
Val Epoch 3: 100%|██████████| 96/96 [00:00<00:00, 416.83it/s, loss_val=0.909, acc_val=0.814]
Epoch 3, Validation Accuracy: 0.8138
Train Epoch 4: 100%|██████████| 382/382 [00:01<00:00, 212.34it/s, loss_train=0.894, acc_train=0.819]
Val Epoch 4: 100%|██████████| 96/96 [00:00<00:00, 458.85it/s, loss_val=1.13, acc_val=0.84]
Epoch 4, Validation Accuracy: 0.8398
Train Epoch 5: 100%|██████████| 382/382 [00:01<00:00, 242.50it/s, loss_train=0.575, acc_train=0.833]
Val Epoch 5: 100%|██████████| 96/96 [00:00<00:00, 343.87it/s, loss_val=0.547, acc_val=0.835]
Epoch 5, Validation Accuracy: 0.8346
Train Epoch 6: 100%|██████████| 382/382 [00:01<00:00, 201.48it/s, loss_train=0.516, acc_train=0.833]
Val Epoch 6: 100%|██████████| 96/96 [00:00<00:00, 312.83it/s, loss_val=0.594, acc_val=0.826]
Epoch 6, Validation Accuracy: 0.8255
Train Epoch 7: 100%|██████████| 382/382 [00:01<00:00, 217.38it/s, loss_train=0.495, acc_train=0.835]
Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 288.91it/s, loss_val=0.621, acc_val=0.833]
Epoch 7, Validation Accuracy: 0.8333
Train Epoch 8: 100%|██████████| 382/382 [00:01<00:00, 204.60it/s, loss_train=0.514, acc_train=0.834]
Val Epoch 8: 100%|██████████| 96/96 [00:00<00:00, 367.25it/s, loss_val=0.933, acc_val=0.832]
Epoch 8, Validation Accuracy: 0.8320
Train Epoch 9: 100%|██████████| 382/382 [00:01<00:00, 214.74it/s, loss_train=0.455, acc_train=0.836]
Val Epoch 9: 100%|██████████| 96/96 [00:00<00:00, 339.48it/s, loss_val=0.448, acc_val=0.832]
Epoch 9, Validation Accuracy: 0.8320
Train Epoch 10: 100%|██████████| 382/382 [00:01<00:00, 210.11it/s, loss_train=0.494, acc_train=0.835]
Val Epoch 10: 100%|██████████| 96/96 [00:00<00:00, 398.64it/s, loss_val=0.426, acc_val=0.833]
Epoch 10, Validation Accuracy: 0.8333

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.83475
acc_val   0.83333
loss_train 0.49358
loss_val   0.42644

```

View run **B1N2L1D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/e7ykh021

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192115-e7ykh021/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192140-mqqv9xq5

Syncing run **B1N2L2D1** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/mqqv9xq5

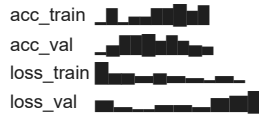
```

Train Epoch 1: 100%|██████████| 382/382 [00:01<00:00, 222.25it/s, loss_train=0.368, acc_train=0.845]
Val Epoch 1: 100%|██████████| 96/96 [00:00<00:00, 407.12it/s, loss_val=0.355, acc_val=0.836]
Epoch 1, Validation Accuracy: 0.8359
Train Epoch 2: 100%|██████████| 382/382 [00:01<00:00, 201.05it/s, loss_train=0.349, acc_train=0.852]
Val Epoch 2: 100%|██████████| 96/96 [00:00<00:00, 348.98it/s, loss_val=0.34, acc_val=0.845]
Epoch 2, Validation Accuracy: 0.8451
Train Epoch 3: 100%|██████████| 382/382 [00:01<00:00, 219.02it/s, loss_train=0.35, acc_train=0.846]
Val Epoch 3: 100%|██████████| 96/96 [00:00<00:00, 270.06it/s, loss_val=0.338, acc_val=0.858]
Epoch 3, Validation Accuracy: 0.8581
Train Epoch 4: 100%|██████████| 382/382 [00:01<00:00, 201.33it/s, loss_train=0.345, acc_train=0.848]
Val Epoch 4: 100%|██████████| 96/96 [00:00<00:00, 305.35it/s, loss_val=0.338, acc_val=0.857]
Epoch 4, Validation Accuracy: 0.8574
Train Epoch 5: 100%|██████████| 382/382 [00:01<00:00, 210.68it/s, loss_train=0.348, acc_train=0.848]
Val Epoch 5: 100%|██████████| 96/96 [00:00<00:00, 407.61it/s, loss_val=0.347, acc_val=0.86]
Epoch 5, Validation Accuracy: 0.8600
Train Epoch 6: 100%|██████████| 382/382 [00:01<00:00, 228.76it/s, loss_train=0.343, acc_train=0.852]
Val Epoch 6: 100%|██████████| 96/96 [00:00<00:00, 310.70it/s, loss_val=0.349, acc_val=0.851]
Epoch 6, Validation Accuracy: 0.8509
Train Epoch 7: 100%|██████████| 382/382 [00:01<00:00, 212.70it/s, loss_train=0.341, acc_train=0.852]
Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 402.43it/s, loss_val=0.341, acc_val=0.855]
Epoch 7, Validation Accuracy: 0.8555
Train Epoch 8: 100%|██████████| 382/382 [00:01<00:00, 233.61it/s, loss_train=0.336, acc_train=0.854]
Val Epoch 8: 100%|██████████| 96/96 [00:00<00:00, 409.57it/s, loss_val=0.363, acc_val=0.851]
Epoch 8, Validation Accuracy: 0.8509
Train Epoch 9: 100%|██████████| 382/382 [00:01<00:00, 221.45it/s, loss_train=0.345, acc_train=0.851]

```

```
Val Epoch 9: 100%|██████████| 96/96 [00:00<00:00, 291.65it/s, loss_val=0.367, acc_val=0.847]
Epoch 9, Validation Accuracy: 0.8470
Train Epoch 10: 100%|██████████| 382/382 [00:01<00:00, 220.40it/s, loss_train=0.337, acc_train=0.853]
Val Epoch 10: 100%|██████████| 96/96 [00:00<00:00, 420.65it/s, loss_val=0.372, acc_val=0.844]Epoch 10, Validation Accuracy: 0.8438
```

Run history:



Run summary:

```
acc_train 0.85308
acc_val 0.84375
loss_train 0.33652
loss_val 0.37226
```

View run **B1N2L2D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/mqqv9xq5
View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing
Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: ./wandb/run-20250326_192140-mqqv9xq5/logs
Tracking run with wandb version 0.19.8
Run data is saved locally in /content/wandb/run-20250326_192206-7tfzd0uz
Syncing run **B1N2L2D2** to [Weights & Biases \(docs\)](#)
View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing
View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/7tfzd0uz
Train Epoch 1: 100%|██████████| 382/382 [00:01<00:00, 243.28it/s, loss_train=0.366, acc_train=0.841]
Val Epoch 1: 100%|██████████| 96/96 [00:00<00:00, 425.14it/s, loss_val=0.354, acc_val=0.837]
Epoch 1, Validation Accuracy: 0.8366
Train Epoch 2: 100%|██████████| 382/382 [00:01<00:00, 214.61it/s, loss_train=0.35, acc_train=0.849]
Val Epoch 2: 100%|██████████| 96/96 [00:00<00:00, 425.77it/s, loss_val=0.353, acc_val=0.857]
Epoch 2, Validation Accuracy: 0.8568
Train Epoch 3: 100%|██████████| 382/382 [00:01<00:00, 240.03it/s, loss_train=0.343, acc_train=0.851]
Val Epoch 3: 100%|██████████| 96/96 [00:00<00:00, 336.05it/s, loss_val=0.335, acc_val=0.855]
Epoch 3, Validation Accuracy: 0.8555
Train Epoch 4: 100%|██████████| 382/382 [00:01<00:00, 211.05it/s, loss_train=0.339, acc_train=0.857]
Val Epoch 4: 100%|██████████| 96/96 [00:00<00:00, 427.77it/s, loss_val=0.361, acc_val=0.85]
Epoch 4, Validation Accuracy: 0.8503
Train Epoch 5: 100%|██████████| 382/382 [00:01<00:00, 210.31it/s, loss_train=0.344, acc_train=0.853]
Val Epoch 5: 100%|██████████| 96/96 [00:00<00:00, 404.53it/s, loss_val=0.383, acc_val=0.837]
Epoch 5, Validation Accuracy: 0.8366
Train Epoch 6: 100%|██████████| 382/382 [00:01<00:00, 227.00it/s, loss_train=0.341, acc_train=0.852]
Val Epoch 6: 100%|██████████| 96/96 [00:00<00:00, 311.03it/s, loss_val=0.341, acc_val=0.852]
Epoch 6, Validation Accuracy: 0.8516
Train Epoch 7: 100%|██████████| 382/382 [00:01<00:00, 211.43it/s, loss_train=0.337, acc_train=0.855]
Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 248.83it/s, loss_val=0.378, acc_val=0.839]
Epoch 7, Validation Accuracy: 0.8392
Train Epoch 8: 100%|██████████| 382/382 [00:01<00:00, 210.59it/s, loss_train=0.333, acc_train=0.856]
Val Epoch 8: 100%|██████████| 96/96 [00:00<00:00, 319.08it/s, loss_val=0.348, acc_val=0.854]
Epoch 8, Validation Accuracy: 0.8535
Train Epoch 9: 100%|██████████| 382/382 [00:01<00:00, 246.32it/s, loss_train=0.334, acc_train=0.855]
Val Epoch 9: 100%|██████████| 96/96 [00:00<00:00, 441.72it/s, loss_val=0.344, acc_val=0.84]
Epoch 9, Validation Accuracy: 0.8398
Train Epoch 10: 100%|██████████| 382/382 [00:01<00:00, 228.44it/s, loss_train=0.334, acc_train=0.852]
Val Epoch 10: 100%|██████████| 96/96 [00:00<00:00, 318.89it/s, loss_val=0.363, acc_val=0.839]
Epoch 10, Validation Accuracy: 0.8392

Run history:



Run summary:

```
acc_train 0.85226
acc_val 0.83919
loss_train 0.33404
loss_val 0.36286
```

View run **B1N2L2D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/7tfzd0uz
 View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing
 Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)
 Find logs at: ./wandb/run-20250326_192206-7tfzd0uz/logs
 Tracking run with wandb version 0.19.8
 Run data is saved locally in /content/wandb/run-20250326_192232-au6q2s1e
 Syncing run **B1N2L3D1** to [Weights & Biases \(docs\)](#)
 View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing
 View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/au6q2s1e

Train Epoch 1: 100%|██████████| 382/382 [00:01<00:00, 229.64it/s, loss_train=0.35, acc_train=0.848]
 Val Epoch 1: 100%|██████████| 96/96 [00:00<00:00, 293.55it/s, loss_val=0.333, acc_val=0.856]
 Epoch 1, Validation Accuracy: 0.8561
 Train Epoch 2: 100%|██████████| 382/382 [00:01<00:00, 233.45it/s, loss_train=0.336, acc_train=0.856]
 Val Epoch 2: 100%|██████████| 96/96 [00:00<00:00, 285.82it/s, loss_val=0.348, acc_val=0.842]
 Epoch 2, Validation Accuracy: 0.8418
 Train Epoch 3: 100%|██████████| 382/382 [00:01<00:00, 192.22it/s, loss_train=0.336, acc_train=0.854]
 Val Epoch 3: 100%|██████████| 96/96 [00:00<00:00, 433.55it/s, loss_val=0.341, acc_val=0.851]
 Epoch 3, Validation Accuracy: 0.8509
 Train Epoch 4: 100%|██████████| 382/382 [00:01<00:00, 232.08it/s, loss_train=0.335, acc_train=0.858]
 Val Epoch 4: 100%|██████████| 96/96 [00:00<00:00, 387.03it/s, loss_val=0.34, acc_val=0.846]
 Epoch 4, Validation Accuracy: 0.8457
 Train Epoch 5: 100%|██████████| 382/382 [00:01<00:00, 221.30it/s, loss_train=0.33, acc_train=0.857]
 Val Epoch 5: 100%|██████████| 96/96 [00:00<00:00, 388.05it/s, loss_val=0.346, acc_val=0.854]
 Epoch 5, Validation Accuracy: 0.8535
 Train Epoch 6: 100%|██████████| 382/382 [00:01<00:00, 224.01it/s, loss_train=0.332, acc_train=0.855]
 Val Epoch 6: 100%|██████████| 96/96 [00:00<00:00, 388.32it/s, loss_val=0.336, acc_val=0.856]
 Epoch 6, Validation Accuracy: 0.8561
 Train Epoch 7: 100%|██████████| 382/382 [00:01<00:00, 211.31it/s, loss_train=0.329, acc_train=0.855]
 Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 316.44it/s, loss_val=0.346, acc_val=0.844]
 Epoch 7, Validation Accuracy: 0.8444
 Train Epoch 8: 100%|██████████| 382/382 [00:01<00:00, 217.27it/s, loss_train=0.329, acc_train=0.854]
 Val Epoch 8: 100%|██████████| 96/96 [00:00<00:00, 405.01it/s, loss_val=0.336, acc_val=0.851]
 Epoch 8, Validation Accuracy: 0.8509
 Train Epoch 9: 100%|██████████| 382/382 [00:01<00:00, 206.42it/s, loss_train=0.326, acc_train=0.856]
 Val Epoch 9: 100%|██████████| 96/96 [00:00<00:00, 346.62it/s, loss_val=0.337, acc_val=0.854]
 Epoch 9, Validation Accuracy: 0.8542
 Train Epoch 10: 100%|██████████| 382/382 [00:01<00:00, 208.83it/s, loss_train=0.329, acc_train=0.857]
 Val Epoch 10: 100%|██████████| 96/96 [00:00<00:00, 450.48it/s, loss_val=0.331, acc_val=0.852]
 Epoch 10, Validation Accuracy: 0.8516

Run history:

acc_train ██████████
 acc_val ██████████
 loss_train ██████████
 loss_val ██████████

Run summary:

acc_train 0.85668
 acc_val 0.85156
 loss_train 0.32939
 loss_val 0.33123

View run **B1N2L3D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/au6q2s1e
 View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing
 Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)
 Find logs at: ./wandb/run-20250326_192232-au6q2s1e/logs
 Tracking run with wandb version 0.19.8
 Run data is saved locally in /content/wandb/run-20250326_192259-fm34ysra
 Syncing run **B1N2L3D2** to [Weights & Biases \(docs\)](#)
 View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing
 View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/fm34ysra

Train Epoch 1: 100%|██████████| 382/382 [00:01<00:00, 235.59it/s, loss_train=0.345, acc_train=0.849]
 Val Epoch 1: 100%|██████████| 96/96 [00:00<00:00, 322.28it/s, loss_val=0.365, acc_val=0.85]
 Epoch 1, Validation Accuracy: 0.8503
 Train Epoch 2: 100%|██████████| 382/382 [00:01<00:00, 225.13it/s, loss_train=0.333, acc_train=0.855]
 Val Epoch 2: 100%|██████████| 96/96 [00:00<00:00, 349.18it/s, loss_val=0.339, acc_val=0.86]
 Epoch 2, Validation Accuracy: 0.8600
 Train Epoch 3: 100%|██████████| 382/382 [00:01<00:00, 215.35it/s, loss_train=0.333, acc_train=0.854]
 Val Epoch 3: 100%|██████████| 96/96 [00:00<00:00, 309.12it/s, loss_val=0.335, acc_val=0.852]
 Epoch 3, Validation Accuracy: 0.8516
 Train Epoch 4: 100%|██████████| 382/382 [00:01<00:00, 203.27it/s, loss_train=0.33, acc_train=0.857]
 Val Epoch 4: 100%|██████████| 96/96 [00:00<00:00, 381.01it/s, loss_val=0.345, acc_val=0.851]
 Epoch 4, Validation Accuracy: 0.8509
 Train Epoch 5: 100%|██████████| 382/382 [00:01<00:00, 224.06it/s, loss_train=0.329, acc_train=0.856]
 Val Epoch 5: 100%|██████████| 96/96 [00:00<00:00, 426.53it/s, loss_val=0.336, acc_val=0.855]
 Epoch 5, Validation Accuracy: 0.8555


```
Train Epoch 6: 100%|██████████| 382/382 [00:01<00:00, 213.50it/s, loss_train=0.329, acc_train=0.857]
Val Epoch 6: 100%|██████████| 96/96 [00:00<00:00, 428.37it/s, loss_val=0.336, acc_val=0.851]
Epoch 6, Validation Accuracy: 0.8509
Train Epoch 7: 100%|██████████| 382/382 [00:01<00:00, 208.32it/s, loss_train=0.323, acc_train=0.858]
Val Epoch 7: 100%|██████████| 96/96 [00:00<00:00, 390.78it/s, loss_val=0.356, acc_val=0.841]
Epoch 7, Validation Accuracy: 0.8411
Train Epoch 8: 100%|██████████| 382/382 [00:01<00:00, 232.52it/s, loss_train=0.325, acc_train=0.855]
Val Epoch 8: 100%|██████████| 96/96 [00:00<00:00, 408.97it/s, loss_val=0.351, acc_val=0.847]
Epoch 8, Validation Accuracy: 0.8470
Train Epoch 9: 100%|██████████| 382/382 [00:01<00:00, 208.45it/s, loss_train=0.326, acc_train=0.858]
Val Epoch 9: 100%|██████████| 96/96 [00:00<00:00, 375.55it/s, loss_val=0.332, acc_val=0.852]
Epoch 9, Validation Accuracy: 0.8522
Train Epoch 10: 100%|██████████| 382/382 [00:01<00:00, 225.29it/s, loss_train=0.321, acc_train=0.858]
Val Epoch 10: 100%|██████████| 96/96 [00:00<00:00, 478.85it/s, loss_val=0.34, acc_val=0.852]
Epoch 10, Validation Accuracy: 0.8516
```

Run history:

acc_train ██████████
acc_val ██████████
loss_train ██████████
loss_val ██████████

Run summary:

acc_train 0.85831
acc_val 0.85156
loss_train 0.32133
loss_val 0.33954

View run **B1N2L3D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/fm34ysra

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192259-fm34ysra/logs

```
# second 12
train_model(batch_size=24, learning_rate=0.1, num_hidden=128, drop_rate=0.3, run_name="B2N1L1D1")
train_model(batch_size=24, learning_rate=0.1, num_hidden=128, drop_rate=0.2, run_name="B2N1L1D2")
train_model(batch_size=24, learning_rate=0.01, num_hidden=128, drop_rate=0.3, run_name="B2N1L2D1")
train_model(batch_size=24, learning_rate=0.01, num_hidden=128, drop_rate=0.2, run_name="B2N1L2D2")
train_model(batch_size=24, learning_rate=0.005, num_hidden=128, drop_rate=0.3, run_name="B2N1L3D1")
train_model(batch_size=24, learning_rate=0.005, num_hidden=128, drop_rate=0.2, run_name="B2N1L3D2")
train_model(batch_size=24, learning_rate=0.1, num_hidden=256, drop_rate=0.3, run_name="B2N2L1D1")
train_model(batch_size=24, learning_rate=0.1, num_hidden=256, drop_rate=0.2, run_name="B2N2L1D2")
train_model(batch_size=24, learning_rate=0.01, num_hidden=256, drop_rate=0.3, run_name="B2N2L2D1")
train_model(batch_size=24, learning_rate=0.01, num_hidden=256, drop_rate=0.2, run_name="B2N2L2D2")
train_model(batch_size=24, learning_rate=0.005, num_hidden=256, drop_rate=0.3, run_name="B2N2L3D1")
train_model(batch_size=24, learning_rate=0.005, num_hidden=256, drop_rate=0.2, run_name="B2N2L3D2")
```


Tracking run with wandb version 0.19.8
 Run data is saved locally in /content/wandb/run-20250326_192442-tt0fs6tt
 Syncing run **B2N1L1D1** to [Weights & Biases \(docs\)](#)
 View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing
 View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/tt0fs6tt

```

Train Epoch 1: 100%|██████████| 255/255 [00:00<00:00, 266.19it/s, loss_train=0.903, acc_train=0.807]
Val Epoch 1: 100%|██████████| 64/64 [00:00<00:00, 379.49it/s, loss_val=0.512, acc_val=0.831]
Epoch 1, Validation Accuracy: 0.8311
Train Epoch 2: 100%|██████████| 255/255 [00:01<00:00, 237.42it/s, loss_train=0.466, acc_train=0.829]
Val Epoch 2: 100%|██████████| 64/64 [00:00<00:00, 395.28it/s, loss_val=0.45, acc_val=0.828]
Epoch 2, Validation Accuracy: 0.8285
Train Epoch 3: 100%|██████████| 255/255 [00:01<00:00, 248.64it/s, loss_train=0.437, acc_train=0.831]
Val Epoch 3: 100%|██████████| 64/64 [00:00<00:00, 436.97it/s, loss_val=0.497, acc_val=0.832]
Epoch 3, Validation Accuracy: 0.8317
Train Epoch 4: 100%|██████████| 255/255 [00:01<00:00, 254.22it/s, loss_train=0.465, acc_train=0.833]
Val Epoch 4: 100%|██████████| 64/64 [00:00<00:00, 386.62it/s, loss_val=0.576, acc_val=0.834]
Epoch 4, Validation Accuracy: 0.8337
Train Epoch 5: 100%|██████████| 255/255 [00:01<00:00, 240.70it/s, loss_train=0.461, acc_train=0.833]
Val Epoch 5: 100%|██████████| 64/64 [00:00<00:00, 393.87it/s, loss_val=0.594, acc_val=0.833]
Epoch 5, Validation Accuracy: 0.8330
Train Epoch 6: 100%|██████████| 255/255 [00:01<00:00, 247.97it/s, loss_train=0.512, acc_train=0.831]
Val Epoch 6: 100%|██████████| 64/64 [00:00<00:00, 393.03it/s, loss_val=0.451, acc_val=0.833]
Epoch 6, Validation Accuracy: 0.8330
Train Epoch 7: 100%|██████████| 255/255 [00:01<00:00, 210.52it/s, loss_train=0.458, acc_train=0.836]
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 275.40it/s, loss_val=0.439, acc_val=0.834]
Epoch 7, Validation Accuracy: 0.8337
Train Epoch 8: 100%|██████████| 255/255 [00:01<00:00, 214.78it/s, loss_train=0.516, acc_train=0.836]
Val Epoch 8: 100%|██████████| 64/64 [00:00<00:00, 325.86it/s, loss_val=0.49, acc_val=0.82]
Epoch 8, Validation Accuracy: 0.8200
Train Epoch 9: 100%|██████████| 255/255 [00:01<00:00, 211.16it/s, loss_train=0.451, acc_train=0.835]
Val Epoch 9: 100%|██████████| 64/64 [00:00<00:00, 360.07it/s, loss_val=0.615, acc_val=0.834]
Epoch 9, Validation Accuracy: 0.8337
Train Epoch 10: 100%|██████████| 255/255 [00:01<00:00, 217.86it/s, loss_train=0.461, acc_train=0.837]
Val Epoch 10: 100%|██████████| 64/64 [00:00<00:00, 372.11it/s, loss_val=0.447, acc_val=0.834]Epoch 10, Validation Accuracy: 0.8337

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.83701
acc_val   0.83366
loss_train 0.46066
loss_val   0.44734

```

View run **B2N1L1D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/tt0fs6tt
 View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing
 Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)
 Find logs at: ./wandb/run-20250326_192442-tt0fs6tt/logs
 Tracking run with wandb version 0.19.8
 Run data is saved locally in /content/wandb/run-20250326_192458-jved07i5
 Syncing run **B2N1L1D2** to [Weights & Biases \(docs\)](#)
 View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing
 View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/jved07i5

```

Train Epoch 1: 100%|██████████| 255/255 [00:00<00:00, 257.77it/s, loss_train=0.638, acc_train=0.818]
Val Epoch 1: 100%|██████████| 64/64 [00:00<00:00, 355.48it/s, loss_val=0.514, acc_val=0.826]
Epoch 1, Validation Accuracy: 0.8265
Train Epoch 2: 100%|██████████| 255/255 [00:01<00:00, 231.48it/s, loss_train=0.574, acc_train=0.823]
Val Epoch 2: 100%|██████████| 64/64 [00:00<00:00, 396.72it/s, loss_val=0.415, acc_val=0.832]
Epoch 2, Validation Accuracy: 0.8324
Train Epoch 3: 100%|██████████| 255/255 [00:01<00:00, 251.88it/s, loss_train=0.465, acc_train=0.836]
Val Epoch 3: 100%|██████████| 64/64 [00:00<00:00, 398.48it/s, loss_val=0.678, acc_val=0.836]
Epoch 3, Validation Accuracy: 0.8363
Train Epoch 4: 100%|██████████| 255/255 [00:01<00:00, 231.98it/s, loss_train=0.456, acc_train=0.834]
Val Epoch 4: 100%|██████████| 64/64 [00:00<00:00, 381.65it/s, loss_val=0.492, acc_val=0.834]
Epoch 4, Validation Accuracy: 0.8337
Train Epoch 5: 100%|██████████| 255/255 [00:01<00:00, 234.75it/s, loss_train=0.427, acc_train=0.836]
Val Epoch 5: 100%|██████████| 64/64 [00:00<00:00, 379.65it/s, loss_val=0.409, acc_val=0.833]
Epoch 5, Validation Accuracy: 0.8330
Train Epoch 6: 100%|██████████| 255/255 [00:01<00:00, 217.74it/s, loss_train=0.42, acc_train=0.836]
Val Epoch 6: 100%|██████████| 64/64 [00:00<00:00, 447.35it/s, loss_val=0.486, acc_val=0.834]
Epoch 6, Validation Accuracy: 0.8337
Train Epoch 7: 100%|██████████| 255/255 [00:01<00:00, 218.16it/s, loss_train=0.503, acc_train=0.833]
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 380.07it/s, loss_val=0.477, acc_val=0.834]
Epoch 7, Validation Accuracy: 0.8337

```

```
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 303.02it/s, loss_val=0.559, acc_val=0.833]
Epoch 7, Validation Accuracy: 0.8343
Train Epoch 8: 100%|██████████| 255/255 [00:01<00:00, 222.61it/s, loss_train=0.41, acc_train=0.838]
Val Epoch 8: 100%|██████████| 64/64 [00:00<00:00, 318.48it/s, loss_val=0.559, acc_val=0.833]
Epoch 8, Validation Accuracy: 0.8330
Train Epoch 9: 100%|██████████| 255/255 [00:01<00:00, 203.04it/s, loss_train=0.45, acc_train=0.837]
Val Epoch 9: 100%|██████████| 64/64 [00:00<00:00, 310.60it/s, loss_val=0.45, acc_val=0.834]
Epoch 9, Validation Accuracy: 0.8337
Train Epoch 10: 100%|██████████| 255/255 [00:01<00:00, 210.04it/s, loss_train=0.435, acc_train=0.839]
Val Epoch 10: 100%|██████████| 64/64 [00:00<00:00, 418.10it/s, loss_val=0.502, acc_val=0.826]
Epoch 10, Validation Accuracy: 0.8258
```

Run history:



acc_train ██████████
acc_val ██████████
loss_train ██████████
loss_val ██████████

Run summary:

acc_train 0.83856
acc_val 0.82585
loss_train 0.43459
loss_val 0.50206

View run **B2N1L1D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/jved07i5

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192458-jved07i5/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192514-3ua33tm7

Syncing run **B2N1L2D1** to [Weights & Biases](https://wandb.ai/farminmansk-fiit-stu/NS_Testing) ([docs](#))

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/3ua33tm7

```
Train Epoch 1: 100%|██████████| 255/255 [00:01<00:00, 249.77it/s, loss_train=0.348, acc_train=0.849]
Val Epoch 1: 100%|██████████| 64/64 [00:00<00:00, 382.89it/s, loss_val=0.344, acc_val=0.854]
Epoch 1, Validation Accuracy: 0.8535
Train Epoch 2: 100%|██████████| 255/255 [00:01<00:00, 224.66it/s, loss_train=0.337, acc_train=0.852]
Val Epoch 2: 100%|██████████| 64/64 [00:00<00:00, 332.93it/s, loss_val=0.336, acc_val=0.847]
Epoch 2, Validation Accuracy: 0.8467
Train Epoch 3: 100%|██████████| 255/255 [00:01<00:00, 234.06it/s, loss_train=0.337, acc_train=0.855]
Val Epoch 3: 100%|██████████| 64/64 [00:00<00:00, 401.28it/s, loss_val=0.344, acc_val=0.85]
Epoch 3, Validation Accuracy: 0.8496
Train Epoch 4: 100%|██████████| 255/255 [00:01<00:00, 225.01it/s, loss_train=0.337, acc_train=0.854]
Val Epoch 4: 100%|██████████| 64/64 [00:00<00:00, 343.70it/s, loss_val=0.362, acc_val=0.835]
Epoch 4, Validation Accuracy: 0.8350
Train Epoch 5: 100%|██████████| 255/255 [00:01<00:00, 247.34it/s, loss_train=0.338, acc_train=0.853]
Val Epoch 5: 100%|██████████| 64/64 [00:00<00:00, 357.72it/s, loss_val=0.335, acc_val=0.846]
Epoch 5, Validation Accuracy: 0.8457
Train Epoch 6: 100%|██████████| 255/255 [00:01<00:00, 217.10it/s, loss_train=0.336, acc_train=0.857]
Val Epoch 6: 100%|██████████| 64/64 [00:00<00:00, 364.31it/s, loss_val=0.331, acc_val=0.855]
Epoch 6, Validation Accuracy: 0.8555
Train Epoch 7: 100%|██████████| 255/255 [00:01<00:00, 211.06it/s, loss_train=0.334, acc_train=0.857]
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 378.29it/s, loss_val=0.337, acc_val=0.854]
Epoch 7, Validation Accuracy: 0.8542
Train Epoch 8: 100%|██████████| 255/255 [00:01<00:00, 177.57it/s, loss_train=0.332, acc_train=0.854]
Val Epoch 8: 100%|██████████| 64/64 [00:00<00:00, 275.36it/s, loss_val=0.339, acc_val=0.855]
Epoch 8, Validation Accuracy: 0.8548
Train Epoch 9: 100%|██████████| 255/255 [00:01<00:00, 199.59it/s, loss_train=0.334, acc_train=0.856]
Val Epoch 9: 100%|██████████| 64/64 [00:00<00:00, 361.79it/s, loss_val=0.352, acc_val=0.852]
Epoch 9, Validation Accuracy: 0.8516
Train Epoch 10: 100%|██████████| 255/255 [00:01<00:00, 242.04it/s, loss_train=0.335, acc_train=0.854]
Val Epoch 10: 100%|██████████| 64/64 [00:00<00:00, 394.92it/s, loss_val=0.337, acc_val=0.844]
Epoch 10, Validation Accuracy: 0.8438
```

Run history:



acc_train ██████████
acc_val ██████████
loss_train ██████████
loss_val ██████████

Run summary:

```
acc_train 0.85392
acc_val   0.84375
loss_train 0.33485
loss_val   0.33686
```

View run **B2N1L2D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/3ua33tm7

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192514-3ua33tm7/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192531-c9yciy6p

Syncing run **B2N1L2D2** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/c9yciy6p

```
Train Epoch 1: 100%|██████████| 255/255 [00:01<00:00, 242.30it/s, loss_train=0.35, acc_train=0.85]
Val Epoch 1: 100%|██████████| 64/64 [00:00<00:00, 373.92it/s, loss_val=0.336, acc_val=0.857]
Epoch 1, Validation Accuracy: 0.8568
Train Epoch 2: 100%|██████████| 255/255 [00:01<00:00, 226.15it/s, loss_train=0.338, acc_train=0.852]
Val Epoch 2: 100%|██████████| 64/64 [00:00<00:00, 380.86it/s, loss_val=0.334, acc_val=0.86]
Epoch 2, Validation Accuracy: 0.8597
Train Epoch 3: 100%|██████████| 255/255 [00:01<00:00, 232.05it/s, loss_train=0.335, acc_train=0.853]
Val Epoch 3: 100%|██████████| 64/64 [00:00<00:00, 428.14it/s, loss_val=0.335, acc_val=0.855]
Epoch 3, Validation Accuracy: 0.8555
Train Epoch 4: 100%|██████████| 255/255 [00:01<00:00, 226.27it/s, loss_train=0.336, acc_train=0.856]
Val Epoch 4: 100%|██████████| 64/64 [00:00<00:00, 299.87it/s, loss_val=0.335, acc_val=0.852]
Epoch 4, Validation Accuracy: 0.8519
Train Epoch 5: 100%|██████████| 255/255 [00:01<00:00, 232.54it/s, loss_train=0.335, acc_train=0.853]
Val Epoch 5: 100%|██████████| 64/64 [00:00<00:00, 405.62it/s, loss_val=0.333, acc_val=0.853]
Epoch 5, Validation Accuracy: 0.8529
Train Epoch 6: 100%|██████████| 255/255 [00:01<00:00, 248.54it/s, loss_train=0.332, acc_train=0.855]
Val Epoch 6: 100%|██████████| 64/64 [00:00<00:00, 327.50it/s, loss_val=0.326, acc_val=0.86]
Epoch 6, Validation Accuracy: 0.8597
Train Epoch 7: 100%|██████████| 255/255 [00:01<00:00, 207.25it/s, loss_train=0.33, acc_train=0.851]
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 322.72it/s, loss_val=0.341, acc_val=0.854]
Epoch 7, Validation Accuracy: 0.8542
Train Epoch 8: 100%|██████████| 255/255 [00:01<00:00, 197.69it/s, loss_train=0.326, acc_train=0.858]
Val Epoch 8: 100%|██████████| 64/64 [00:00<00:00, 329.23it/s, loss_val=0.341, acc_val=0.851]
Epoch 8, Validation Accuracy: 0.8509
Train Epoch 9: 100%|██████████| 255/255 [00:00<00:00, 269.87it/s, loss_train=0.327, acc_train=0.859]
Val Epoch 9: 100%|██████████| 64/64 [00:00<00:00, 487.76it/s, loss_val=0.378, acc_val=0.833]
Epoch 9, Validation Accuracy: 0.8327
Train Epoch 10: 100%|██████████| 255/255 [00:01<00:00, 219.47it/s, loss_train=0.332, acc_train=0.857]
Val Epoch 10: 100%|██████████| 64/64 [00:00<00:00, 402.40it/s, loss_val=0.332, acc_val=0.859]
Epoch 10, Validation Accuracy: 0.8590
```

Run history:

```
acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val   ██████████
```

Run summary:

```
acc_train 0.85711
acc_val   0.85905
loss_train 0.3322
loss_val   0.33159
```

View run **B2N1L2D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/c9yciy6p

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192531-c9yciy6p/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192547-qp21iqum

Syncing run **B2N1L3D1** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/qp21iqum

```
Train Epoch 1: 100%|██████████| 255/255 [00:01<00:00, 247.50it/s, loss_train=0.344, acc_train=0.849]
Val Epoch 1: 100%|██████████| 64/64 [00:00<00:00, 343.31it/s, loss_val=0.343, acc_val=0.85]
Epoch 1, Validation Accuracy: 0.8503
Train Epoch 2: 100%|██████████| 255/255 [00:01<00:00, 224.65it/s, loss_train=0.331, acc_train=0.857]
Val Epoch 2: 100%|██████████| 64/64 [00:00<00:00, 407.64it/s, loss_val=0.327, acc_val=0.858]
Epoch 2, Validation Accuracy: 0.8584
Train Epoch 3: 100%|██████████| 255/255 [00:01<00:00, 230.04it/s, loss_train=0.328, acc_train=0.861]
Val Epoch 3: 100%|██████████| 64/64 [00:00<00:00, 307.78it/s, loss_val=0.326, acc_val=0.861]
Epoch 3, Validation Accuracy: 0.8610
Train Epoch 4: 100%|██████████| 255/255 [00:01<00:00, 229.09it/s, loss_train=0.329, acc_train=0.856]
```

```
Val Epoch 4: 100%|██████████| 64/64 [00:00<00:00, 332.91it/s, loss_val=0.329, acc_val=0.854]
Epoch 4, Validation Accuracy: 0.8535
Train Epoch 5: 100%|██████████| 255/255 [00:01<00:00, 220.55it/s, loss_train=0.323, acc_train=0.857]
Val Epoch 5: 100%|██████████| 64/64 [00:00<00:00, 462.59it/s, loss_val=0.337, acc_val=0.85]
Epoch 5, Validation Accuracy: 0.8503
Train Epoch 6: 100%|██████████| 255/255 [00:01<00:00, 216.05it/s, loss_train=0.327, acc_train=0.853]
Val Epoch 6: 100%|██████████| 64/64 [00:00<00:00, 440.42it/s, loss_val=0.329, acc_val=0.852]
Epoch 6, Validation Accuracy: 0.8516
Train Epoch 7: 100%|██████████| 255/255 [00:01<00:00, 229.62it/s, loss_train=0.323, acc_train=0.861]
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 249.76it/s, loss_val=0.343, acc_val=0.851]
Epoch 7, Validation Accuracy: 0.8512
Train Epoch 8: 100%|██████████| 255/255 [00:01<00:00, 215.45it/s, loss_train=0.327, acc_train=0.858]
Val Epoch 8: 100%|██████████| 64/64 [00:00<00:00, 374.17it/s, loss_val=0.331, acc_val=0.848]
Epoch 8, Validation Accuracy: 0.8483
Train Epoch 9: 100%|██████████| 255/255 [00:01<00:00, 214.88it/s, loss_train=0.325, acc_train=0.86]
Val Epoch 9: 100%|██████████| 64/64 [00:00<00:00, 329.73it/s, loss_val=0.33, acc_val=0.85]
Epoch 9, Validation Accuracy: 0.8503
Train Epoch 10: 100%|██████████| 255/255 [00:01<00:00, 204.52it/s, loss_train=0.323, acc_train=0.855]
Val Epoch 10: 100%|██████████| 64/64 [00:00<00:00, 368.75it/s, loss_val=0.334, acc_val=0.855]
Epoch 10, Validation Accuracy: 0.8548
```

Run history:

acc_train ██████████
acc_val ██████████
loss_train ██████████
loss_val ██████████

Run summary:

acc_train 0.85498
acc_val 0.85482
loss_train 0.32316
loss_val 0.3342

View run **B2N1L3D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/qp21iqum

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192547-qp21iqum/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192603-jdu9m1nd

Syncing run **B2N1L3D2** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/jdu9m1nd

```
Train Epoch 1: 100%|██████████| 255/255 [00:01<00:00, 237.45it/s, loss_train=0.345, acc_train=0.854]
Val Epoch 1: 100%|██████████| 64/64 [00:00<00:00, 394.88it/s, loss_val=0.347, acc_val=0.838]
Epoch 1, Validation Accuracy: 0.8379
Train Epoch 2: 100%|██████████| 255/255 [00:01<00:00, 219.95it/s, loss_train=0.331, acc_train=0.857]
Val Epoch 2: 100%|██████████| 64/64 [00:00<00:00, 341.79it/s, loss_val=0.337, acc_val=0.852]
Epoch 2, Validation Accuracy: 0.8519
Train Epoch 3: 100%|██████████| 255/255 [00:01<00:00, 226.12it/s, loss_train=0.325, acc_train=0.863]
Val Epoch 3: 100%|██████████| 64/64 [00:00<00:00, 404.62it/s, loss_val=0.336, acc_val=0.857]
Epoch 3, Validation Accuracy: 0.8568
Train Epoch 4: 100%|██████████| 255/255 [00:01<00:00, 201.98it/s, loss_train=0.326, acc_train=0.858]
Val Epoch 4: 100%|██████████| 64/64 [00:00<00:00, 337.38it/s, loss_val=0.329, acc_val=0.856]
Epoch 4, Validation Accuracy: 0.8564
Train Epoch 5: 100%|██████████| 255/255 [00:01<00:00, 221.46it/s, loss_train=0.325, acc_train=0.856]
Val Epoch 5: 100%|██████████| 64/64 [00:00<00:00, 443.44it/s, loss_val=0.33, acc_val=0.849]
Epoch 5, Validation Accuracy: 0.8490
Train Epoch 6: 100%|██████████| 255/255 [00:01<00:00, 232.98it/s, loss_train=0.324, acc_train=0.859]
Val Epoch 6: 100%|██████████| 64/64 [00:00<00:00, 422.77it/s, loss_val=0.346, acc_val=0.848]
Epoch 6, Validation Accuracy: 0.8483
Train Epoch 7: 100%|██████████| 255/255 [00:01<00:00, 229.41it/s, loss_train=0.323, acc_train=0.861]
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 314.67it/s, loss_val=0.334, acc_val=0.845]
Epoch 7, Validation Accuracy: 0.8454
Train Epoch 8: 100%|██████████| 255/255 [00:01<00:00, 187.96it/s, loss_train=0.321, acc_train=0.858]
Val Epoch 8: 100%|██████████| 64/64 [00:00<00:00, 470.52it/s, loss_val=0.329, acc_val=0.854]
Epoch 8, Validation Accuracy: 0.8542
Train Epoch 9: 100%|██████████| 255/255 [00:01<00:00, 196.12it/s, loss_train=0.321, acc_train=0.861]
Val Epoch 9: 100%|██████████| 64/64 [00:00<00:00, 423.63it/s, loss_val=0.33, acc_val=0.852]
Epoch 9, Validation Accuracy: 0.8516
Train Epoch 10: 100%|██████████| 255/255 [00:01<00:00, 240.60it/s, loss_train=0.319, acc_train=0.859]
Val Epoch 10: 100%|██████████| 64/64 [00:00<00:00, 384.37it/s, loss_val=0.327, acc_val=0.853]Epoch 10, Validation Accuracy: 0.8532
```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.85948
acc_val   0.85319
loss_train 0.31868
loss_val   0.32707

```

View run **B2N1L3D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/jdu9m1nd

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192603-jdu9m1nd/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192619-18u5v9oo

Syncing run **B2N2L1D1** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/18u5v9oo

```

Train Epoch 1: 100%|██████████| 255/255 [00:01<00:00, 246.60it/s, loss_train=1.16, acc_train=0.809]
Val Epoch 1: 100%|██████████| 64/64 [00:00<00:00, 324.66it/s, loss_val=0.862, acc_val=0.806]
Epoch 1, Validation Accuracy: 0.8057
Train Epoch 2: 100%|██████████| 255/255 [00:01<00:00, 206.64it/s, loss_train=0.823, acc_train=0.817]
Val Epoch 2: 100%|██████████| 64/64 [00:00<00:00, 469.78it/s, loss_val=0.876, acc_val=0.8]
Epoch 2, Validation Accuracy: 0.8001
Train Epoch 3: 100%|██████████| 255/255 [00:01<00:00, 229.41it/s, loss_train=0.623, acc_train=0.825]
Val Epoch 3: 100%|██████████| 64/64 [00:00<00:00, 409.92it/s, loss_val=0.78, acc_val=0.819]
Epoch 3, Validation Accuracy: 0.8193
Train Epoch 4: 100%|██████████| 255/255 [00:01<00:00, 239.91it/s, loss_train=0.616, acc_train=0.823]
Val Epoch 4: 100%|██████████| 64/64 [00:00<00:00, 395.52it/s, loss_val=0.547, acc_val=0.826]
Epoch 4, Validation Accuracy: 0.8258
Train Epoch 5: 100%|██████████| 255/255 [00:01<00:00, 214.55it/s, loss_train=0.563, acc_train=0.829]
Val Epoch 5: 100%|██████████| 64/64 [00:00<00:00, 330.67it/s, loss_val=0.757, acc_val=0.819]
Epoch 5, Validation Accuracy: 0.8193
Train Epoch 6: 100%|██████████| 255/255 [00:01<00:00, 216.50it/s, loss_train=0.687, acc_train=0.826]
Val Epoch 6: 100%|██████████| 64/64 [00:00<00:00, 408.25it/s, loss_val=0.693, acc_val=0.832]
Epoch 6, Validation Accuracy: 0.8324
Train Epoch 7: 100%|██████████| 255/255 [00:01<00:00, 219.67it/s, loss_train=0.564, acc_train=0.831]
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 276.92it/s, loss_val=0.475, acc_val=0.834]
Epoch 7, Validation Accuracy: 0.8337
Train Epoch 8: 100%|██████████| 255/255 [00:01<00:00, 184.51it/s, loss_train=0.493, acc_train=0.834]
Val Epoch 8: 100%|██████████| 64/64 [00:00<00:00, 354.06it/s, loss_val=0.592, acc_val=0.834]
Epoch 8, Validation Accuracy: 0.8337
Train Epoch 9: 100%|██████████| 255/255 [00:01<00:00, 213.99it/s, loss_train=0.524, acc_train=0.834]
Val Epoch 9: 100%|██████████| 64/64 [00:00<00:00, 301.54it/s, loss_val=0.571, acc_val=0.834]
Epoch 9, Validation Accuracy: 0.8337
Train Epoch 10: 100%|██████████| 255/255 [00:01<00:00, 212.89it/s, loss_train=0.475, acc_train=0.836]
Val Epoch 10: 100%|██████████| 64/64 [00:00<00:00, 472.59it/s, loss_val=0.479, acc_val=0.832]
Epoch 10, Validation Accuracy: 0.8324

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.8357
acc_val   0.83236
loss_train 0.47502
loss_val   0.47907

```

View run **B2N2L1D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/18u5v9oo

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192619-18u5v9oo/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192636-25urbc9q

Syncing run **B2N2L1D2** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/25urbc9q

```

Train Epoch 1: 100%|██████████| 255/255 [00:01<00:00, 238.93it/s, loss_train=1.19, acc_train=0.811]
Val Epoch 1: 100%|██████████| 64/64 [00:00<00:00, 443.91it/s, loss_val=0.824, acc_val=0.818]
Epoch 1, Validation Accuracy: 0.8177
Train Epoch 2: 100%|██████████| 255/255 [00:01<00:00, 232.82it/s, loss_train=0.667, acc_train=0.821]
Val Epoch 2: 100%|██████████| 64/64 [00:00<00:00, 255.61it/s, loss_val=0.512, acc_val=0.82]
Epoch 2, Validation Accuracy: 0.8197
Train Epoch 3: 100%|██████████| 255/255 [00:01<00:00, 220.82it/s, loss_train=0.476, acc_train=0.83]
Val Epoch 3: 100%|██████████| 64/64 [00:00<00:00, 361.83it/s, loss_val=0.512, acc_val=0.812]
Epoch 3, Validation Accuracy: 0.8115
Train Epoch 4: 100%|██████████| 255/255 [00:01<00:00, 205.22it/s, loss_train=0.496, acc_train=0.828]
Val Epoch 4: 100%|██████████| 64/64 [00:00<00:00, 376.35it/s, loss_val=1.46, acc_val=0.831]
Epoch 4, Validation Accuracy: 0.8311
Train Epoch 5: 100%|██████████| 255/255 [00:01<00:00, 209.75it/s, loss_train=0.565, acc_train=0.832]
Val Epoch 5: 100%|██████████| 64/64 [00:00<00:00, 324.98it/s, loss_val=0.917, acc_val=0.834]
Epoch 5, Validation Accuracy: 0.8337
Train Epoch 6: 100%|██████████| 255/255 [00:01<00:00, 193.93it/s, loss_train=0.575, acc_train=0.83]
Val Epoch 6: 100%|██████████| 64/64 [00:00<00:00, 296.40it/s, loss_val=0.561, acc_val=0.828]
Epoch 6, Validation Accuracy: 0.8278
Train Epoch 7: 100%|██████████| 255/255 [00:01<00:00, 207.92it/s, loss_train=0.591, acc_train=0.828]
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 457.06it/s, loss_val=0.485, acc_val=0.834]
Epoch 7, Validation Accuracy: 0.8337
Train Epoch 8: 100%|██████████| 255/255 [00:01<00:00, 208.42it/s, loss_train=0.438, acc_train=0.837]
Val Epoch 8: 100%|██████████| 64/64 [00:00<00:00, 297.06it/s, loss_val=0.538, acc_val=0.824]
Epoch 8, Validation Accuracy: 0.8239
Train Epoch 9: 100%|██████████| 255/255 [00:01<00:00, 193.71it/s, loss_train=0.452, acc_train=0.835]
Val Epoch 9: 100%|██████████| 64/64 [00:00<00:00, 338.48it/s, loss_val=0.45, acc_val=0.82]
Epoch 9, Validation Accuracy: 0.8200
Train Epoch 10: 100%|██████████| 255/255 [00:01<00:00, 210.69it/s, loss_train=0.522, acc_train=0.835]
Val Epoch 10: 100%|██████████| 64/64 [00:00<00:00, 360.82it/s, loss_val=0.628, acc_val=0.838]
Epoch 10, Validation Accuracy: 0.8379

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.83489
acc_val   0.83789
loss_train 0.5217
loss_val   0.62757

```

View run **B2N2L1D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/25urbc9q

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192636-25urbc9q/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192653-ua4ejlhj

Syncing run **B2N2L2D1** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/ua4ejlhj

```

Train Epoch 1: 100%|██████████| 255/255 [00:01<00:00, 246.80it/s, loss_train=0.358, acc_train=0.845]
Val Epoch 1: 100%|██████████| 64/64 [00:00<00:00, 404.96it/s, loss_val=0.348, acc_val=0.837]
Epoch 1, Validation Accuracy: 0.8372
Train Epoch 2: 100%|██████████| 255/255 [00:01<00:00, 212.84it/s, loss_train=0.343, acc_train=0.855]
Val Epoch 2: 100%|██████████| 64/64 [00:00<00:00, 293.29it/s, loss_val=0.344, acc_val=0.854]
Epoch 2, Validation Accuracy: 0.8542
Train Epoch 3: 100%|██████████| 255/255 [00:01<00:00, 218.91it/s, loss_train=0.343, acc_train=0.85]
Val Epoch 3: 100%|██████████| 64/64 [00:00<00:00, 381.90it/s, loss_val=0.336, acc_val=0.852]
Epoch 3, Validation Accuracy: 0.8519
Train Epoch 4: 100%|██████████| 255/255 [00:01<00:00, 245.87it/s, loss_train=0.338, acc_train=0.854]
Val Epoch 4: 100%|██████████| 64/64 [00:00<00:00, 309.73it/s, loss_val=0.352, acc_val=0.851]
Epoch 4, Validation Accuracy: 0.8506
Train Epoch 5: 100%|██████████| 255/255 [00:01<00:00, 191.20it/s, loss_train=0.338, acc_train=0.855]
Val Epoch 5: 100%|██████████| 64/64 [00:00<00:00, 298.76it/s, loss_val=0.331, acc_val=0.853]
Epoch 5, Validation Accuracy: 0.8525
Train Epoch 6: 100%|██████████| 255/255 [00:01<00:00, 201.57it/s, loss_train=0.337, acc_train=0.854]
Val Epoch 6: 100%|██████████| 64/64 [00:00<00:00, 435.59it/s, loss_val=0.374, acc_val=0.841]
Epoch 6, Validation Accuracy: 0.8415
Train Epoch 7: 100%|██████████| 255/255 [00:01<00:00, 193.19it/s, loss_train=0.338, acc_train=0.851]
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 367.79it/s, loss_val=0.356, acc_val=0.848]
Epoch 7, Validation Accuracy: 0.8480
Train Epoch 8: 100%|██████████| 255/255 [00:01<00:00, 205.55it/s, loss_train=0.339, acc_train=0.853]
Val Epoch 8: 100%|██████████| 64/64 [00:00<00:00, 377.78it/s, loss_val=0.336, acc_val=0.85]
Epoch 8, Validation Accuracy: 0.8503
Train Epoch 9: 100%|██████████| 255/255 [00:01<00:00, 239.71it/s, loss_train=0.34, acc_train=0.854]

```



```
Val Epoch 9: 100%|██████████| 64/64 [00:00<00:00, 391.16it/s, loss_val=0.345, acc_val=0.848]
Epoch 9, Validation Accuracy: 0.8480
Train Epoch 10: 100%|██████████| 255/255 [00:01<00:00, 189.59it/s, loss_train=0.337, acc_train=0.852]
Val Epoch 10: 100%|██████████| 64/64 [00:00<00:00, 298.13it/s, loss_val=0.336, acc_val=0.859]
Epoch 10, Validation Accuracy: 0.8590
```

Run history:

acc_train ██████████
acc_val ██████████
loss_train ██████████
loss_val ██████████

Run summary:

acc_train 0.85229
acc_val 0.85905
loss_train 0.33703
loss_val 0.33648

View run **B2N2L2D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/ua4ejlhj

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192653-ua4ejlhj/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_192711-fzkmrqfu

Syncing run **B2N2L2D2** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/fzkmrqfu

```
Train Epoch 1: 100%|██████████| 255/255 [00:01<00:00, 216.85it/s, loss_train=0.356, acc_train=0.85]
Val Epoch 1: 100%|██████████| 64/64 [00:00<00:00, 415.22it/s, loss_val=0.341, acc_val=0.855]
Epoch 1, Validation Accuracy: 0.8555
Train Epoch 2: 100%|██████████| 255/255 [00:01<00:00, 224.37it/s, loss_train=0.337, acc_train=0.857]
Val Epoch 2: 100%|██████████| 64/64 [00:00<00:00, 431.46it/s, loss_val=0.357, acc_val=0.836]
Epoch 2, Validation Accuracy: 0.8359
Train Epoch 3: 100%|██████████| 255/255 [00:01<00:00, 208.57it/s, loss_train=0.341, acc_train=0.856]
Val Epoch 3: 100%|██████████| 64/64 [00:00<00:00, 468.24it/s, loss_val=0.348, acc_val=0.842]
Epoch 3, Validation Accuracy: 0.8421
Train Epoch 4: 100%|██████████| 255/255 [00:01<00:00, 207.88it/s, loss_train=0.336, acc_train=0.851]
Val Epoch 4: 100%|██████████| 64/64 [00:00<00:00, 333.17it/s, loss_val=0.338, acc_val=0.855]
Epoch 4, Validation Accuracy: 0.8555
Train Epoch 5: 100%|██████████| 255/255 [00:01<00:00, 213.51it/s, loss_train=0.337, acc_train=0.856]
Val Epoch 5: 100%|██████████| 64/64 [00:00<00:00, 395.44it/s, loss_val=0.358, acc_val=0.842]
Epoch 5, Validation Accuracy: 0.8424
Train Epoch 6: 100%|██████████| 255/255 [00:01<00:00, 232.05it/s, loss_train=0.339, acc_train=0.853]
Val Epoch 6: 100%|██████████| 64/64 [00:00<00:00, 402.99it/s, loss_val=0.339, acc_val=0.852]
Epoch 6, Validation Accuracy: 0.8519
Train Epoch 7: 100%|██████████| 255/255 [00:01<00:00, 197.19it/s, loss_train=0.331, acc_train=0.859]
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 287.63it/s, loss_val=0.347, acc_val=0.85]
Epoch 7, Validation Accuracy: 0.8496
Train Epoch 8: 100%|██████████| 255/255 [00:01<00:00, 191.73it/s, loss_train=0.331, acc_train=0.855]
Val Epoch 8: 100%|██████████| 64/64 [00:00<00:00, 230.74it/s, loss_val=0.356, acc_val=0.848]
Epoch 8, Validation Accuracy: 0.8480
Train Epoch 9: 100%|██████████| 255/255 [00:01<00:00, 189.29it/s, loss_train=0.333, acc_train=0.852]
Val Epoch 9: 100%|██████████| 64/64 [00:00<00:00, 421.71it/s, loss_val=0.341, acc_val=0.852]
Epoch 9, Validation Accuracy: 0.8522
Train Epoch 10: 100%|██████████| 255/255 [00:01<00:00, 234.39it/s, loss_train=0.327, acc_train=0.858]
Val Epoch 10: 100%|██████████| 64/64 [00:00<00:00, 400.06it/s, loss_val=0.361, acc_val=0.838]
Epoch 10, Validation Accuracy: 0.8382
```

Run history:

acc_train ██████████
acc_val ██████████
loss_train ██████████
loss_val ██████████

Run summary:

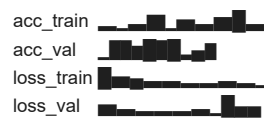
acc_train 0.85784
acc_val 0.83822
loss_train 0.32694
loss_val 0.36079

View run **B2N2L2D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/fzkmrqfu
 View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing
 Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)
 Find logs at: ./wandb/run-20250326_192711-fzkmrqfu/logs
 Tracking run with wandb version 0.19.8
 Run data is saved locally in /content/wandb/run-20250326_192729-4ztdbhpX
 Syncing run **B2N2L3D1** to [Weights & Biases \(docs\)](#)
 View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing
 View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/4ztdbhpX

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	0.347	0.854	0.345	0.842
2	0.338	0.853	0.337	0.854
3	0.333	0.855	0.335	0.854
4	0.33	0.858	0.337	0.85
5	0.33	0.853	0.336	0.855
6	0.326	0.855	0.338	0.853
7	0.326	0.854	0.332	0.855
8	0.328	0.857	0.354	0.844
9	0.326	0.858	0.341	0.847
10	0.323	0.854	0.341	0.852

Epoch 10, Validation Accuracy: 0.8516

Run history:



Run summary:

acc_train 0.85425
 acc_val 0.85156
 loss_train 0.32291
 loss_val 0.34059

View run **B2N2L3D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/4ztdbhpX
 View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing
 Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)
 Find logs at: ./wandb/run-20250326_192729-4ztdbhpX/logs
 Tracking run with wandb version 0.19.8
 Run data is saved locally in /content/wandb/run-20250326_192748-a460ppnm
 Syncing run **B2N2L3D2** to [Weights & Biases \(docs\)](#)
 View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing
 View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/a460ppnm

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	0.345	0.851	0.346	0.857
2	0.329	0.861	0.339	0.853
3	0.329	0.86	0.336	0.856
4	0.326	0.863	0.336	0.85
5	0.328	0.86	0.338	0.849

Epoch 5, Validation Accuracy: 0.8490


```

Train Epoch 6: 100%|██████████| 255/255 [00:01<00:00, 207.27it/s, loss_train=0.324, acc_train=0.856]
Val Epoch 6: 100%|██████████| 64/64 [00:00<00:00, 334.04it/s, loss_val=0.347, acc_val=0.836]
Epoch 6, Validation Accuracy: 0.8359
Train Epoch 7: 100%|██████████| 255/255 [00:01<00:00, 200.74it/s, loss_train=0.321, acc_train=0.861]
Val Epoch 7: 100%|██████████| 64/64 [00:00<00:00, 276.26it/s, loss_val=0.333, acc_val=0.85]
Epoch 7, Validation Accuracy: 0.8499
Train Epoch 8: 100%|██████████| 255/255 [00:01<00:00, 208.11it/s, loss_train=0.321, acc_train=0.859]
Val Epoch 8: 100%|██████████| 64/64 [00:00<00:00, 383.01it/s, loss_val=0.334, acc_val=0.852]
Epoch 8, Validation Accuracy: 0.8522
Train Epoch 9: 100%|██████████| 255/255 [00:01<00:00, 195.56it/s, loss_train=0.321, acc_train=0.86]
Val Epoch 9: 100%|██████████| 64/64 [00:00<00:00, 287.18it/s, loss_val=0.342, acc_val=0.85]
Epoch 9, Validation Accuracy: 0.8499
Train Epoch 10: 100%|██████████| 255/255 [00:01<00:00, 186.47it/s, loss_train=0.319, acc_train=0.861]
Val Epoch 10: 100%|██████████| 64/64 [00:00<00:00, 399.43it/s, loss_val=0.33, acc_val=0.858]
Epoch 10, Validation Accuracy: 0.8577

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.86111
acc_val   0.85775
loss_train 0.31876
loss_val   0.33019

```

View run **B2N2L3D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/a460ppnm

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_192748-a460ppnm/logs

```
# last 12
train_model(batch_size=32, learning_rate=0.1, num_hidden=128, drop_rate=0.3, run_name="B3N1L1D1")
train_model(batch_size=32, learning_rate=0.1, num_hidden=128, drop_rate=0.2, run_name="B3N1L1D2")
train_model(batch_size=32, learning_rate=0.01, num_hidden=128, drop_rate=0.3, run_name="B3N1L2D1")
train_model(batch_size=32, learning_rate=0.01, num_hidden=128, drop_rate=0.2, run_name="B3N1L2D2")
train_model(batch_size=32, learning_rate=0.005, num_hidden=128, drop_rate=0.3, run_name="B3N1L3D1")
train_model(batch_size=32, learning_rate=0.005, num_hidden=128, drop_rate=0.2, run_name="B3N1L3D2")
train_model(batch_size=32, learning_rate=0.1, num_hidden=256, drop_rate=0.3, run_name="B3N2L1D1")
train_model(batch_size=32, learning_rate=0.1, num_hidden=256, drop_rate=0.2, run_name="B3N2L1D2")
train_model(batch_size=32, learning_rate=0.01, num_hidden=256, drop_rate=0.3, run_name="B3N2L2D1")
train_model(batch_size=32, learning_rate=0.01, num_hidden=256, drop_rate=0.2, run_name="B3N2L2D2")
train_model(batch_size=32, learning_rate=0.005, num_hidden=256, drop_rate=0.3, run_name="B3N2L3D1")
train_model(batch_size=32, learning_rate=0.005, num_hidden=256, drop_rate=0.2, run_name="B3N2L3D2")
```



Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_193121-tf0or81e

Syncing run **B3N1L1D1** to [Weights & Biases \(docs\)](#)View project at https://wandb.ai/farminmansk-fiit-stu/NS_TestingView run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/tf0or81e

```

Train Epoch 1: 100%|██████████| 191/191 [00:00<00:00, 247.60it/s, loss_train=0.633, acc_train=0.818]
Val Epoch 1: 100%|██████████| 48/48 [00:00<00:00, 409.43it/s, loss_val=0.46, acc_val=0.827]
Epoch 1, Validation Accuracy: 0.8268
Train Epoch 2: 100%|██████████| 191/191 [00:00<00:00, 247.85it/s, loss_train=0.467, acc_train=0.827]
Val Epoch 2: 100%|██████████| 48/48 [00:00<00:00, 370.52it/s, loss_val=0.447, acc_val=0.832]
Epoch 2, Validation Accuracy: 0.8320
Train Epoch 3: 100%|██████████| 191/191 [00:00<00:00, 233.34it/s, loss_train=0.469, acc_train=0.829]
Val Epoch 3: 100%|██████████| 48/48 [00:00<00:00, 442.56it/s, loss_val=0.63, acc_val=0.827]
Epoch 3, Validation Accuracy: 0.8268
Train Epoch 4: 100%|██████████| 191/191 [00:00<00:00, 261.13it/s, loss_train=0.507, acc_train=0.833]
Val Epoch 4: 100%|██████████| 48/48 [00:00<00:00, 399.10it/s, loss_val=0.428, acc_val=0.822]
Epoch 4, Validation Accuracy: 0.8223
Train Epoch 5: 100%|██████████| 191/191 [00:00<00:00, 226.66it/s, loss_train=0.445, acc_train=0.832]
Val Epoch 5: 100%|██████████| 48/48 [00:00<00:00, 387.55it/s, loss_val=0.488, acc_val=0.833]
Epoch 5, Validation Accuracy: 0.8333
Train Epoch 6: 100%|██████████| 191/191 [00:00<00:00, 243.36it/s, loss_train=0.447, acc_train=0.833]
Val Epoch 6: 100%|██████████| 48/48 [00:00<00:00, 348.59it/s, loss_val=0.446, acc_val=0.83]
Epoch 6, Validation Accuracy: 0.8301
Train Epoch 7: 100%|██████████| 191/191 [00:00<00:00, 219.60it/s, loss_train=0.426, acc_train=0.838]
Val Epoch 7: 100%|██████████| 48/48 [00:00<00:00, 356.14it/s, loss_val=0.432, acc_val=0.833]
Epoch 7, Validation Accuracy: 0.8333
Train Epoch 8: 100%|██████████| 191/191 [00:00<00:00, 202.86it/s, loss_train=0.41, acc_train=0.837]
Val Epoch 8: 100%|██████████| 48/48 [00:00<00:00, 415.80it/s, loss_val=0.465, acc_val=0.833]
Epoch 8, Validation Accuracy: 0.8333
Train Epoch 9: 100%|██████████| 191/191 [00:00<00:00, 233.10it/s, loss_train=0.43, acc_train=0.837]
Val Epoch 9: 100%|██████████| 48/48 [00:00<00:00, 342.20it/s, loss_val=0.519, acc_val=0.833]
Epoch 9, Validation Accuracy: 0.8327
Train Epoch 10: 100%|██████████| 191/191 [00:00<00:00, 194.96it/s, loss_train=0.451, acc_train=0.836]
Val Epoch 10: 100%|██████████| 48/48 [00:00<00:00, 301.46it/s, loss_val=0.451, acc_val=0.833]
Epoch 10, Validation Accuracy: 0.8333

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.83606
acc_val   0.83333
loss_train 0.4511
loss_val   0.45092

```

View run **B3N1L1D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/tf0or81eView project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_193121-tf0or81e/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_193134-4b5kns12

Syncing run **B3N1L1D2** to [Weights & Biases \(docs\)](#)View project at https://wandb.ai/farminmansk-fiit-stu/NS_TestingView run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/4b5kns12

```

Train Epoch 1: 100%|██████████| 191/191 [00:00<00:00, 243.10it/s, loss_train=0.606, acc_train=0.821]
Val Epoch 1: 100%|██████████| 48/48 [00:00<00:00, 428.26it/s, loss_val=0.5, acc_val=0.827]
Epoch 1, Validation Accuracy: 0.8268
Train Epoch 2: 100%|██████████| 191/191 [00:00<00:00, 235.88it/s, loss_train=0.482, acc_train=0.83]
Val Epoch 2: 100%|██████████| 48/48 [00:00<00:00, 341.81it/s, loss_val=0.7, acc_val=0.806]
Epoch 2, Validation Accuracy: 0.8056
Train Epoch 3: 100%|██████████| 191/191 [00:00<00:00, 254.12it/s, loss_train=0.435, acc_train=0.835]
Val Epoch 3: 100%|██████████| 48/48 [00:00<00:00, 336.73it/s, loss_val=0.46, acc_val=0.833]
Epoch 3, Validation Accuracy: 0.8333
Train Epoch 4: 100%|██████████| 191/191 [00:00<00:00, 228.92it/s, loss_train=0.426, acc_train=0.832]
Val Epoch 4: 100%|██████████| 48/48 [00:00<00:00, 421.65it/s, loss_val=0.404, acc_val=0.833]
Epoch 4, Validation Accuracy: 0.8333
Train Epoch 5: 100%|██████████| 191/191 [00:00<00:00, 261.66it/s, loss_train=0.446, acc_train=0.834]
Val Epoch 5: 100%|██████████| 48/48 [00:00<00:00, 327.79it/s, loss_val=0.414, acc_val=0.831]
Epoch 5, Validation Accuracy: 0.8314
Train Epoch 6: 100%|██████████| 191/191 [00:00<00:00, 219.55it/s, loss_train=0.456, acc_train=0.831]
Val Epoch 6: 100%|██████████| 48/48 [00:00<00:00, 341.94it/s, loss_val=0.413, acc_val=0.831]
Epoch 6, Validation Accuracy: 0.8307
Train Epoch 7: 100%|██████████| 191/191 [00:00<00:00, 225.46it/s, loss_train=0.436, acc_train=0.836]
Val Epoch 7: 100%|██████████| 48/48 [00:00<00:00, 351.18it/s, loss_val=0.417, acc_val=0.831]

```

```

Val Epoch 7: 100%|██████████| 48/48 [00:00<00:00, 321.20it/s, loss_val=0.447, acc_val=0.833]
Epoch 7, Validation Accuracy: 0.8314
Train Epoch 8: 100%|██████████| 191/191 [00:00<00:00, 241.93it/s, loss_train=0.418, acc_train=0.839]
Val Epoch 8: 100%|██████████| 48/48 [00:00<00:00, 342.54it/s, loss_val=0.447, acc_val=0.833]
Epoch 8, Validation Accuracy: 0.8333
Train Epoch 9: 100%|██████████| 191/191 [00:00<00:00, 201.27it/s, loss_train=0.412, acc_train=0.838]
Val Epoch 9: 100%|██████████| 48/48 [00:00<00:00, 299.49it/s, loss_val=0.437, acc_val=0.827]
Epoch 9, Validation Accuracy: 0.8275
Train Epoch 10: 100%|██████████| 191/191 [00:00<00:00, 227.69it/s, loss_train=0.436, acc_train=0.837]
Val Epoch 10: 100%|██████████| 48/48 [00:00<00:00, 447.04it/s, loss_val=0.472, acc_val=0.82]
Epoch 10, Validation Accuracy: 0.8197

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.83655
acc_val   0.81966
loss_train 0.43617
loss_val   0.4725

```

View run **B3N1L1D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/4b5kns12

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_193134-4b5kns12/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_193147-81d8ohib

Syncing run **B3N1L2D1** to [Weights & Biases](https://wandb.ai/farminmansk-fiit-stu/NS_Testing) ([docs](#))

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/81d8ohib

```

Train Epoch 1: 100%|██████████| 191/191 [00:00<00:00, 246.78it/s, loss_train=0.354, acc_train=0.848]
Val Epoch 1: 100%|██████████| 48/48 [00:00<00:00, 338.26it/s, loss_val=0.338, acc_val=0.856]
Epoch 1, Validation Accuracy: 0.8559
Train Epoch 2: 100%|██████████| 191/191 [00:00<00:00, 216.16it/s, loss_train=0.333, acc_train=0.855]
Val Epoch 2: 100%|██████████| 48/48 [00:00<00:00, 336.31it/s, loss_val=0.347, acc_val=0.843]
Epoch 2, Validation Accuracy: 0.8429
Train Epoch 3: 100%|██████████| 191/191 [00:00<00:00, 241.56it/s, loss_train=0.335, acc_train=0.851]
Val Epoch 3: 100%|██████████| 48/48 [00:00<00:00, 345.65it/s, loss_val=0.342, acc_val=0.846]
Epoch 3, Validation Accuracy: 0.8464
Train Epoch 4: 100%|██████████| 191/191 [00:00<00:00, 225.67it/s, loss_train=0.335, acc_train=0.854]
Val Epoch 4: 100%|██████████| 48/48 [00:00<00:00, 402.67it/s, loss_val=0.339, acc_val=0.856]
Epoch 4, Validation Accuracy: 0.8557
Train Epoch 5: 100%|██████████| 191/191 [00:00<00:00, 250.04it/s, loss_train=0.335, acc_train=0.853]
Val Epoch 5: 100%|██████████| 48/48 [00:00<00:00, 317.50it/s, loss_val=0.333, acc_val=0.856]
Epoch 5, Validation Accuracy: 0.8557
Train Epoch 6: 100%|██████████| 191/191 [00:00<00:00, 222.94it/s, loss_train=0.332, acc_train=0.855]
Val Epoch 6: 100%|██████████| 48/48 [00:00<00:00, 361.42it/s, loss_val=0.335, acc_val=0.861]
Epoch 6, Validation Accuracy: 0.8613
Train Epoch 7: 100%|██████████| 191/191 [00:00<00:00, 214.52it/s, loss_train=0.332, acc_train=0.852]
Val Epoch 7: 100%|██████████| 48/48 [00:00<00:00, 359.54it/s, loss_val=0.352, acc_val=0.851]
Epoch 7, Validation Accuracy: 0.8511
Train Epoch 8: 100%|██████████| 191/191 [00:00<00:00, 248.53it/s, loss_train=0.33, acc_train=0.856]
Val Epoch 8: 100%|██████████| 48/48 [00:00<00:00, 367.61it/s, loss_val=0.342, acc_val=0.842]
Epoch 8, Validation Accuracy: 0.8420
Train Epoch 9: 100%|██████████| 191/191 [00:00<00:00, 227.51it/s, loss_train=0.335, acc_train=0.852]
Val Epoch 9: 100%|██████████| 48/48 [00:00<00:00, 315.58it/s, loss_val=0.334, acc_val=0.85]
Epoch 9, Validation Accuracy: 0.8496
Train Epoch 10: 100%|██████████| 191/191 [00:01<00:00, 188.05it/s, loss_train=0.334, acc_train=0.855]
Val Epoch 10: 100%|██████████| 48/48 [00:00<00:00, 412.07it/s, loss_val=0.336, acc_val=0.852]
Epoch 10, Validation Accuracy: 0.8516

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```
acc_train 0.85537
acc_val   0.85156
loss_train 0.33355
loss_val   0.33644
```

View run **B3N1L2D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/81d8ohib

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_193147-81d8ohib/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_193200-wk8qz1a2

Syncing run **B3N1L2D2** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/wk8qz1a2

```
Train Epoch 1: 100%|██████████| 191/191 [00:00<00:00, 242.39it/s, loss_train=0.346, acc_train=0.848]
Val Epoch 1: 100%|██████████| 48/48 [00:00<00:00, 379.28it/s, loss_val=0.335, acc_val=0.857]
Epoch 1, Validation Accuracy: 0.8568
Train Epoch 2: 100%|██████████| 191/191 [00:00<00:00, 229.19it/s, loss_train=0.335, acc_train=0.853]
Val Epoch 2: 100%|██████████| 48/48 [00:00<00:00, 428.67it/s, loss_val=0.342, acc_val=0.842]
Epoch 2, Validation Accuracy: 0.8418
Train Epoch 3: 100%|██████████| 191/191 [00:00<00:00, 238.84it/s, loss_train=0.33, acc_train=0.857]
Val Epoch 3: 100%|██████████| 48/48 [00:00<00:00, 333.17it/s, loss_val=0.343, acc_val=0.859]
Epoch 3, Validation Accuracy: 0.8592
Train Epoch 4: 100%|██████████| 191/191 [00:00<00:00, 214.88it/s, loss_train=0.329, acc_train=0.858]
Val Epoch 4: 100%|██████████| 48/48 [00:00<00:00, 302.02it/s, loss_val=0.334, acc_val=0.859]
Epoch 4, Validation Accuracy: 0.8585
Train Epoch 5: 100%|██████████| 191/191 [00:01<00:00, 134.93it/s, loss_train=0.331, acc_train=0.854]
Val Epoch 5: 100%|██████████| 48/48 [00:00<00:00, 380.82it/s, loss_val=0.362, acc_val=0.845]
Epoch 5, Validation Accuracy: 0.8448
Train Epoch 6: 100%|██████████| 191/191 [00:01<00:00, 127.19it/s, loss_train=0.334, acc_train=0.854]
Val Epoch 6: 100%|██████████| 48/48 [00:00<00:00, 391.24it/s, loss_val=0.338, acc_val=0.845]
Epoch 6, Validation Accuracy: 0.8455
Train Epoch 7: 100%|██████████| 191/191 [00:00<00:00, 226.80it/s, loss_train=0.328, acc_train=0.859]
Val Epoch 7: 100%|██████████| 48/48 [00:00<00:00, 403.35it/s, loss_val=0.337, acc_val=0.848]
Epoch 7, Validation Accuracy: 0.8477
Train Epoch 8: 100%|██████████| 191/191 [00:00<00:00, 213.90it/s, loss_train=0.33, acc_train=0.855]
Val Epoch 8: 100%|██████████| 48/48 [00:00<00:00, 407.18it/s, loss_val=0.338, acc_val=0.857]
Epoch 8, Validation Accuracy: 0.8568
Train Epoch 9: 100%|██████████| 191/191 [00:00<00:00, 195.73it/s, loss_train=0.325, acc_train=0.854]
Val Epoch 9: 100%|██████████| 48/48 [00:00<00:00, 365.30it/s, loss_val=0.339, acc_val=0.851]
Epoch 9, Validation Accuracy: 0.8513
Train Epoch 10: 100%|██████████| 191/191 [00:00<00:00, 235.32it/s, loss_train=0.322, acc_train=0.857]
Val Epoch 10: 100%|██████████| 48/48 [00:00<00:00, 306.47it/s, loss_val=0.35, acc_val=0.851]
Epoch 10, Validation Accuracy: 0.8507
```

Run history:

```
acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val   ██████████
```

Run summary:

```
acc_train 0.85733
acc_val   0.85069
loss_train 0.32215
loss_val   0.35038
```

View run **B3N1L2D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/wk8qz1a2

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_193200-wk8qz1a2/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_193215-y094fxeb

Syncing run **B3N1L3D1** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/y094fxeb

```
Train Epoch 1: 100%|██████████| 191/191 [00:00<00:00, 233.58it/s, loss_train=0.346, acc_train=0.846]
Val Epoch 1: 100%|██████████| 48/48 [00:00<00:00, 338.74it/s, loss_val=0.345, acc_val=0.846]
Epoch 1, Validation Accuracy: 0.8464
Train Epoch 2: 100%|██████████| 191/191 [00:00<00:00, 204.13it/s, loss_train=0.332, acc_train=0.857]
Val Epoch 2: 100%|██████████| 48/48 [00:00<00:00, 268.87it/s, loss_val=0.341, acc_val=0.837]
Epoch 2, Validation Accuracy: 0.8366
Train Epoch 3: 100%|██████████| 191/191 [00:01<00:00, 139.08it/s, loss_train=0.328, acc_train=0.856]
Val Epoch 3: 100%|██████████| 48/48 [00:00<00:00, 414.46it/s, loss_val=0.329, acc_val=0.861]
Epoch 3, Validation Accuracy: 0.8609
Train Epoch 4: 100%|██████████| 191/191 [00:00<00:00, 201.98it/s, loss_train=0.33, acc_train=0.855]
```

```

Val Epoch 4: 100%|██████████| 48/48 [00:00<00:00, 328.14it/s, loss_val=0.33, acc_val=0.858]
Epoch 4, Validation Accuracy: 0.8581
Train Epoch 5: 100%|██████████| 191/191 [00:00<00:00, 214.13it/s, loss_train=0.324, acc_train=0.858]
Val Epoch 5: 100%|██████████| 48/48 [00:00<00:00, 387.05it/s, loss_val=0.331, acc_val=0.855]
Epoch 5, Validation Accuracy: 0.8548
Train Epoch 6: 100%|██████████| 191/191 [00:00<00:00, 239.84it/s, loss_train=0.327, acc_train=0.86]
Val Epoch 6: 100%|██████████| 48/48 [00:00<00:00, 382.36it/s, loss_val=0.331, acc_val=0.853]
Epoch 6, Validation Accuracy: 0.8526
Train Epoch 7: 100%|██████████| 191/191 [00:01<00:00, 119.60it/s, loss_train=0.322, acc_train=0.861]
Val Epoch 7: 100%|██████████| 48/48 [00:00<00:00, 390.40it/s, loss_val=0.331, acc_val=0.852]
Epoch 7, Validation Accuracy: 0.8520
Train Epoch 8: 100%|██████████| 191/191 [00:02<00:00, 75.53it/s, loss_train=0.321, acc_train=0.858]
Val Epoch 8: 100%|██████████| 48/48 [00:00<00:00, 120.79it/s, loss_val=0.333, acc_val=0.859]
Epoch 8, Validation Accuracy: 0.8585
Train Epoch 9: 100%|██████████| 191/191 [00:01<00:00, 160.94it/s, loss_train=0.322, acc_train=0.859]
Val Epoch 9: 100%|██████████| 48/48 [00:00<00:00, 287.33it/s, loss_val=0.329, acc_val=0.855]
Epoch 9, Validation Accuracy: 0.8553
Train Epoch 10: 100%|██████████| 191/191 [00:01<00:00, 190.46it/s, loss_train=0.321, acc_train=0.855]
Val Epoch 10: 100%|██████████| 48/48 [00:00<00:00, 296.69it/s, loss_val=0.335, acc_val=0.851]
Epoch 10, Validation Accuracy: 0.8513

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.8552
acc_val   0.85135
loss_train 0.32056
loss_val   0.33489

```

View run **B3N1L3D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/y094fxeb

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_193215-y094fxeb/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_193231-3xngqttp

Syncing run **B3N1L3D2** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/3xngqttp

```

Train Epoch 1: 100%|██████████| 191/191 [00:00<00:00, 255.42it/s, loss_train=0.346, acc_train=0.852]
Val Epoch 1: 100%|██████████| 48/48 [00:00<00:00, 383.37it/s, loss_val=0.334, acc_val=0.854]
Epoch 1, Validation Accuracy: 0.8539
Train Epoch 2: 100%|██████████| 191/191 [00:00<00:00, 209.69it/s, loss_train=0.327, acc_train=0.858]
Val Epoch 2: 100%|██████████| 48/48 [00:00<00:00, 362.86it/s, loss_val=0.337, acc_val=0.848]
Epoch 2, Validation Accuracy: 0.8477
Train Epoch 3: 100%|██████████| 191/191 [00:00<00:00, 211.76it/s, loss_train=0.325, acc_train=0.857]
Val Epoch 3: 100%|██████████| 48/48 [00:00<00:00, 412.03it/s, loss_val=0.327, acc_val=0.859]
Epoch 3, Validation Accuracy: 0.8587
Train Epoch 4: 100%|██████████| 191/191 [00:00<00:00, 227.12it/s, loss_train=0.323, acc_train=0.86]
Val Epoch 4: 100%|██████████| 48/48 [00:00<00:00, 420.93it/s, loss_val=0.336, acc_val=0.853]
Epoch 4, Validation Accuracy: 0.8529
Train Epoch 5: 100%|██████████| 191/191 [00:00<00:00, 226.06it/s, loss_train=0.323, acc_train=0.862]
Val Epoch 5: 100%|██████████| 48/48 [00:00<00:00, 361.42it/s, loss_val=0.338, acc_val=0.854]
Epoch 5, Validation Accuracy: 0.8542
Train Epoch 6: 100%|██████████| 191/191 [00:00<00:00, 232.74it/s, loss_train=0.323, acc_train=0.857]
Val Epoch 6: 100%|██████████| 48/48 [00:00<00:00, 345.96it/s, loss_val=0.332, acc_val=0.855]
Epoch 6, Validation Accuracy: 0.8550
Train Epoch 7: 100%|██████████| 191/191 [00:00<00:00, 211.65it/s, loss_train=0.322, acc_train=0.86]
Val Epoch 7: 100%|██████████| 48/48 [00:00<00:00, 436.62it/s, loss_val=0.327, acc_val=0.854]
Epoch 7, Validation Accuracy: 0.8535
Train Epoch 8: 100%|██████████| 191/191 [00:00<00:00, 238.73it/s, loss_train=0.321, acc_train=0.86]
Val Epoch 8: 100%|██████████| 48/48 [00:00<00:00, 387.77it/s, loss_val=0.328, acc_val=0.856]
Epoch 8, Validation Accuracy: 0.8559
Train Epoch 9: 100%|██████████| 191/191 [00:00<00:00, 223.61it/s, loss_train=0.317, acc_train=0.865]
Val Epoch 9: 100%|██████████| 48/48 [00:00<00:00, 276.99it/s, loss_val=0.327, acc_val=0.858]
Epoch 9, Validation Accuracy: 0.8581
Train Epoch 10: 100%|██████████| 191/191 [00:01<00:00, 180.47it/s, loss_train=0.315, acc_train=0.862]
Val Epoch 10: 100%|██████████| 48/48 [00:00<00:00, 275.16it/s, loss_val=0.331, acc_val=0.848]
Epoch 10, Validation Accuracy: 0.8477

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.86224
acc_val   0.84766
loss_train 0.31498
loss_val  0.33089

```

View run **B3N1L3D2** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/3xnggttq

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_193231-3xnggttq/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_193244-f4qtztn7

Syncing run **B3N2L1D1** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/f4qtztn7

```

Train Epoch 1: 100%|██████████| 191/191 [00:00<00:00, 224.92it/s, loss_train=1.21, acc_train=0.806]
Val Epoch 1: 100%|██████████| 48/48 [00:00<00:00, 413.66it/s, loss_val=0.682, acc_val=0.833]
Epoch 1, Validation Accuracy: 0.8327
Train Epoch 2: 100%|██████████| 191/191 [00:00<00:00, 210.07it/s, loss_train=0.546, acc_train=0.82]
Val Epoch 2: 100%|██████████| 48/48 [00:00<00:00, 316.62it/s, loss_val=0.524, acc_val=0.824]
Epoch 2, Validation Accuracy: 0.8244
Train Epoch 3: 100%|██████████| 191/191 [00:00<00:00, 229.47it/s, loss_train=0.531, acc_train=0.827]
Val Epoch 3: 100%|██████████| 48/48 [00:00<00:00, 447.07it/s, loss_val=0.482, acc_val=0.827]
Epoch 3, Validation Accuracy: 0.8273
Train Epoch 4: 100%|██████████| 191/191 [00:00<00:00, 211.50it/s, loss_train=0.533, acc_train=0.829]
Val Epoch 4: 100%|██████████| 48/48 [00:00<00:00, 441.52it/s, loss_val=0.628, acc_val=0.833]
Epoch 4, Validation Accuracy: 0.8333
Train Epoch 5: 100%|██████████| 191/191 [00:00<00:00, 259.43it/s, loss_train=0.495, acc_train=0.832]
Val Epoch 5: 100%|██████████| 48/48 [00:00<00:00, 468.48it/s, loss_val=0.601, acc_val=0.834]
Epoch 5, Validation Accuracy: 0.8340
Train Epoch 6: 100%|██████████| 191/191 [00:00<00:00, 227.66it/s, loss_train=0.51, acc_train=0.83]
Val Epoch 6: 100%|██████████| 48/48 [00:00<00:00, 435.07it/s, loss_val=0.653, acc_val=0.812]
Epoch 6, Validation Accuracy: 0.8125
Train Epoch 7: 100%|██████████| 191/191 [00:00<00:00, 216.09it/s, loss_train=0.513, acc_train=0.83]
Val Epoch 7: 100%|██████████| 48/48 [00:00<00:00, 444.16it/s, loss_val=0.492, acc_val=0.824]
Epoch 7, Validation Accuracy: 0.8236
Train Epoch 8: 100%|██████████| 191/191 [00:00<00:00, 247.18it/s, loss_train=0.482, acc_train=0.833]
Val Epoch 8: 100%|██████████| 48/48 [00:00<00:00, 270.26it/s, loss_val=0.623, acc_val=0.84]
Epoch 8, Validation Accuracy: 0.8398
Train Epoch 9: 100%|██████████| 191/191 [00:00<00:00, 229.76it/s, loss_train=0.544, acc_train=0.83]
Val Epoch 9: 100%|██████████| 48/48 [00:00<00:00, 406.94it/s, loss_val=0.571, acc_val=0.819]
Epoch 9, Validation Accuracy: 0.8186
Train Epoch 10: 100%|██████████| 191/191 [00:00<00:00, 201.30it/s, loss_train=0.591, acc_train=0.829]
Val Epoch 10: 100%|██████████| 48/48 [00:00<00:00, 313.58it/s, loss_val=0.569, acc_val=0.833]
Epoch 10, Validation Accuracy: 0.8333

```

Run history:

```

acc_train ██████████
acc_val   ██████████
loss_train ██████████
loss_val  ██████████

```

Run summary:

```

acc_train 0.82902
acc_val   0.83333
loss_train 0.59076
loss_val  0.56861

```

View run **B3N2L1D1** at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/f4qtztn7

View project at: https://wandb.ai/farminmansk-fiit-stu/NS_Testing

Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)

Find logs at: ./wandb/run-20250326_193244-f4qtztn7/logs

Tracking run with wandb version 0.19.8

Run data is saved locally in /content/wandb/run-20250326_193257-m4mfekex

Syncing run **B3N2L1D2** to [Weights & Biases \(docs\)](#)

View project at https://wandb.ai/farminmansk-fiit-stu/NS_Testing

View run at https://wandb.ai/farminmansk-fiit-stu/NS_Testing/runs/m4mfekex